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Patient Assessment, Examination, Diagnosis, and Treatment Planning

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This chapter provides an overview of the process through which a clinician completes patient assessment, clinical examination, diagnosis, and treatment planning for operative dentistry procedures. The chapter assumes that the reader has a background in oral medicine and an understanding of how to perform complete (comprehensive) extraoral and intraoral hard and soft tissue examinations, as well as an understanding of the etiology, characteristics, risk assessment, and management of dental caries as presented in Chapter 2. It is not in the scope of this chapter to incorporate the details of other aspects of a complete dental examination, such as periodontal, occlusal, and esthetic examinations. Appropriate textbooks that cover the specifics of these areas, in health and disease, should be consulted.

Any discussion of diagnosis and treatment must begin with an appreciation of the role of the dentist in helping patients maintain their oral health. This role is summarized by the Latin phrase "primum non nocere," which means "do no harm." This phrase represents a fundamental principle continually embraced by those in the healing arts over many centuries.

The implication of this concept for operative dentistry is that, before we recommend treatment, we must be reasonably confident that the patient will be better off as a result of our intervention. However, how can we be reasonably confident when we realize that few, if any, of the tests we perform or the assessments of risk that we make are completely accurate? To make matters even more challenging, none of the treatments we provide is without adverse outcomes and none will likely last for the life of the patient. The answer is that we must acknowledge that the information or evidence we have is not perfect and that we must be clear about the possible consequences of our decisions. If we are informed and clear about options and their consequences, then we reduce the chances of doing any harm.

The success of operative treatment depends heavily on an appropriate plan of care, which, in turn, is based on a comprehensive analysis of the patient's reasons for seeking care and on a systematic assessment of the patient's current conditions and risk for future problems. This information is then combined with the best available evidence on approaches to management of the patient's needs so that an appropriate plan of care may be offered.

The collection of this information and the determinations based on examination findings should be comprehensive and accomplished in a stepwise manner. Simply put, skipping steps may lead to overlooking potentially important parts of the patient's individual needs. These steps include reasons for seeking care, medical and dental histories, clinical examination for the detection of abnormalities, establishing diagnoses (which includes assessing risk), and determining prognosis. All of these steps must occur before a sound and appropriate plan of care may be developed and recommended.

Growing attention to using only the most effective and appropriate treatment has spawned interest in numerous research efforts. Research that provides information on treatments that work best in certain situations is expanding the knowledge base of dentistry and has led to an interest in translating the results of that research into practice activities that enhance care for patients. This movement has been termed *evidence-based dentistry* and is defined as the "conscientious, explicit, and judicious use of current best evidence in making decisions about the care of individual patients."[1] Systematic reviews emerging from the focus on evidence-based dentistry will provide practitioners with a distillation of the available knowledge about various conditions and treatments. It is incumbent on the authors of the systematic review to openly discuss the strengths and weaknesses of the reviewed studies as well as the relative value of their conclusions for application in dental care. Currently, the American Dental Association (ADA) has developed a website (http://ebd.ada.org/) that may be used by dental professionals for evidence-based dentistry decision making. This website helps clinicians identify systematic reviews, describes the preferred method for assembling the best available scientific evidence, and provides an appraisal of the evidence through critical summaries. As evidence-based dentistry continues to expand, professional associations will become more active in the development of guidelines to assist dentists and their patients in making informed and appropriate decisions.

General Considerations

It is difficult to overstate the importance of gaining comprehensive insight into each patient. Dentistry has, by its very origins, been heavily focused on reconstruction of damaged areas. However, nothing that we design and create has the ability to withstand the wet, warm, salty, thermally cycled, and cyclically loaded environment of the oral cavity for the whole life of the patient. Therefore the emphasis in dentistry has shifted toward understanding and maintaining conditions consistent with a healthy stomatognathic

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system so that steps may be taken to prevent dental disease. The specific circumstances of each individual must be considered in light of the known requirements of optimal oral health. Gaining insight into individual circumstances begins with proper patient assessment. Medical and dental health survey questions are excellent basic tools designed to facilitate this process. Responses to the broad overview questions (generally referred to as the "medical history" and "dental history") then enable specific exploration of previous or current conditions unique to the individual patient, that may represent risk factors or indicators for dental disease, as well as the primary reason (i.e., the chief concern) that prompted the patient to seek the assistance of the dentist. This interview process is then followed by the clinical gathering of additional information by means of strategic examination. The examination is the "hands-on" process of observing the patient's extraoral and intraoral structures and detecting of symptoms and signs of abnormal conditions or disease. During the clinical examination, the dentist must be keenly sensitive to subtle symptoms (that the patient reports), signs (that the dentist detects), and variations from normal to detect pathologic conditions and determine etiologic factors. The discovery of additional risk factors/indicators may occur during the examination. The combined patient assessment and examination information is then used to formulate diagnoses (and risk profiles), which are a determination or judgment of health versus disease, variations from normal, and likelihood for the development of additional disease. The dentist must be committed to comprehensive and meticulous attention to detail.

Patient Assessment

Medical History

The patient or legal guardian completes a standard, comprehensive medical history form. This form is an integral part of the preexamination patient interview, which helps identify conditions that could alter, complicate, or contraindicate proposed dental procedures. The practitioner should identify (1) communicable diseases that require special precautions, procedures, or referral; (2) allergies or medications, which may contraindicate the use of certain drugs; (3) systemic diseases, cardiac abnormalities, or joint replacements, which may require prophylactic antibiotic coverage or other treatment modifications; and (4) physiologic changes associated with aging, which may alter clinical presentation and influence treatment. The practitioner also might identify a need for medical consultation or referral before initiating dental care. All of this information is carefully detailed in the patient's permanent record and is used, as needed, to shape subsequent treatment recommendations.

Dental History

The dental history is a review of previous dental experiences and current dental problems. Review of the dental history often reveals information about past dental problems, previous dental treatment, and the patient's responses to treatments. Frequency of dental care and perceptions of previous care may be indications of the patient's future behavior. If a patient has difficulty tolerating certain types of procedures or has encountered problems with previous dental care, an alteration of the treatment or environment might help avoid future complications. It is crucial to understand past experiences in order to provide optimal care in the future. Also, this discussion might lead to identification of specific problems such as areas of food impaction, inability to floss, areas of pain, and broken restorations or tooth structure. Finally, the date, type, and diagnostic quality of available radiographs should be recorded so as to ascertain the need for additional radiographs and to minimize the patient's exposure to ionizing radiation.

Chief Concern

Before initiating any treatment, the patient's chief concerns, or the problems that initiated the patient's visit, should be identified and clearly understood. Concerns are recorded essentially verbatim in the dental record. The patient should be encouraged to discuss all aspects (symptoms) of the current problem(s), including onset, duration, and related factors they are experiencing. This information is vital to establishing which specific diagnostic tests are required, determining the cause, selecting appropriate treatment options for the concerns, and building a sound relationship with the patient.

Examination

It is somewhat artificial to discuss *examination* as a separate entity from patient assessment for aspects of the patient "examination" begin during initial conversations with the patient. Careful observation of extraoral symmetry of the patient's physical appearance of the head and neck areas, mandibular movement during speech, ability to articulate sounds, and tendencies to smile provides vital information relative to overall presence or absence of abnormalities or disease. These observations occur while reviewing/clarifying information reported in the medical and dental history and while listening to the patient's chief concern(s). By definition, these early observations are all extraoral in nature.

Many examination data recording systems utilize organizational logic that begins with "extraoral examination" followed by "intraoral examination" so as to facilitate the recording of observational information (what the dentist observes while interacting with the patient). Utilization of clinical photography to capture full face and profile images is particularly useful in this process. Any observations will ultimately be followed by the physical examination necessary to assess extraoral aspects of the muscles of mastication, temporomandibular joints (TMJs), lymphnodes, and other vital structures, which will then be followed by intraoral examination.

Examination of Esthetic Appearance

Examination of esthetic appearance may be described as the evaluation of tooth color, form, display, and position in relation to the face. Evaluation must include discussion of realistic esthetic expectations when considering treatment options with the patient. Attaining the desired esthetic outcomes may be complicated by maximum tooth display and excessive or uneven tissue display. Risk of patient dissatisfaction with treatment outcomes may be lowered by careful attention to the establishment of intrafacial, intraarch, and interarch tooth positions that have been identified as consistent with maximum esthetics. This is accomplished in light of the reality that when individual teeth are correct in their anatomic shape, and positioned in the face and arches for optimum function, then the overall esthetic result will be optimal ("form follows function"). Tooth color evaluation becomes a factor if teeth are more visible when smiling or at the resting position of lips. Darker colored teeth, teeth with enamel intrinsic staining, and conditions such as tetracycline staining all increase the risk for not satisfying the esthetic expectations of patients with tooth color concerns. Symmetry of gingival margins becomes very important

in patients who display a large amount of gingival tissue when smiling. Lack of symmetry increases the risk of not meeting the patient's esthetic expectations. Presence of multiple risk factors requires in-depth, careful consideration of the various components/relationships of the stomatognathic system, the ability to develop an interdisciplinary treatment plan, and excellent listening skills so as to identify realistic options consistent with the patient's overall esthetic expectations. All of this must be accomplished without compromising the short- and long-term dental health of the patient (*"do no harm"*). In many of these situations, conservative direct or indirect enamel-supported restorations are more appropriate for long-term risk management than are more aggressive preparations that remove relatively more tooth structure.

Examination of Occlusion

A careful examination of the patient's current occlusal scheme, along with potential impact on the muscles of mastication and TMJs, must occur before planning and implementing restorative care (see Chapter 1). This examination includes identification of signs of occlusal trauma, such as heavy wear facets, enamel cracks, or tooth mobility, and notation of occlusal abnormalities that may be contributing to pathologic conditions such as bone loss. Identification of the current relative health of the stomatognathic system then allows consideration of the potential ability of the proposed restorative treatment to achieve harmonious function of each component of the system. Careful analysis may identify need for modification of the current occlusal scheme prior to the initiation of any definitive restorative care.

The static and dynamic occlusion must be examined carefully (see Chapter 1) in light of the observation that there is no "ideal" occlusion and that most patients may have the ability to adapt to their occlusion without clinical symptoms. However, the clinician must understand the normal physiologic response of the muscles of mastication to various occlusal interrelationships and be able to identify where, for a specific patient, pathology (of the dentition, muscles of mastication, and/or TMJs) is present and what modifications may be indicated. A description of the patient's static anatomic occlusion in maximum intercuspation, including the relationship between molars and canines (Angle Class I, II, or III), and the amount of vertical overlap (overbite) and horizontal overlap (overjet) of anterior teeth should be recorded. This should include assessment of the presence and specifics of any functional shift from centric relation occlusion to maximum intercuspation. The presence of missing teeth and the relationship of the maxillary and mandibular midlines should be determined. The appropriateness of the occlusal plane and the positions of malposed teeth should be identified. Supererupted teeth, spacing, fractured teeth, and marginal ridge discrepancies should be noted. The dynamic functional occlusion in all movements of the mandible (right, left, forward, and all excursions in between) should be evaluated. The evaluation also includes assessing the relationship of teeth in centric relation, which is the orthopedic position of the joint where the condyle head is in its most anterior and superior position against the articular eminence within the glenoid fossa. Functional movements of the mandible are evaluated to determine if canine guidance or group function exists. The presence and amount of anterior guidance is evaluated to note the degree of potential posterior disclusion. Teeth are examined for abnormal wear patterns that are excessive and not age appropriate. If signs of abnormal or premature wear are present, the patient is queried as to awareness of any contributing parafunction habits such as grinding or clenching. Nonworking-side excursive contacts are recorded and related to any findings of masticatory muscle myositis and/or ipsilateral TMJ disc issues. Working-side excursive contacts are recorded and related to areas of cusp fracture development. Protrusive contacts on all posterior teeth molars are noted. Heavy wear facets on posterior cuspal inclines, mobility of teeth, or fremitus during function is identified and classified as primary or secondary occlusal traumatism. Full analysis of the occlusion may require articulated diagnostic models. Movement of the mandible from maximum intercuspation to maximum opening is observed and maximum unassisted opening is measured; any "clicking or popping" of the joint disc(s) during mandibular movements is noted and related to any history of trauma, nonworking occlusal interferences, or other possible pathologic changes. Bimanual loading of the joints and palpation of the condyle lateral poles and retrocondylar areas (during wide mandibular opening) are completed to further test for tenderness/pain as signs of inflammation. The occlusal relationships of the teeth are assessed for the presence of an unusually tall and narrow cusp (a "plunger cusp") that "plunges" deep into the occlusal plane of the opposing arch**.** A plunger cusp might contact the lower of two adjacent marginal ridges of different levels, contacting directly between two adjacent marginal ridges in maximum intercuspation, or be positioned in a deep fossa. It may increase the likelihood of food impaction and tooth or restoration fracture.

The results of the occlusal examination should be included in the dental record and considered in the restorative treatment plan. Acceptable aspects of the occlusion must be preserved and not altered during treatment. When possible, improvement of the occlusion (elimination of interferences), based on knowledge of the physiologic masticatory muscle response to various relationships, is desirable; occlusal interferences must not be perpetuated in the restorative treatment.

Examination of Teeth and Restorations

Preparation for Clinical Examination

A trained assistant familiar with the terminology, notation system, and charting procedure may survey the patient's teeth and existing restorations and record the information to save chair time for the dentist. The dentist subsequently performs the examination and confirms the charting. Proper instruments, including a mirror, an explorer, and a periodontal probe, and the ability to air-dry the surfaces of the teeth are required. Every accessible surface of each tooth must be inspected for localized changes in color, texture, and translucency. A routine for charting should be established, such as starting in the upper right quadrant with the most posterior tooth and progressing around the maxillary and mandibular arches. Dental floss is useful in identifying overhanging restorations, improper proximal contours, and open contacts. The clinical examination is performed systematically in a clean, dry, well-illuminated mouth. A cotton roll in the vestibular space and another under the tongue maintain dryness and improve visualization of the teeth and adjacent gingiva (Fig. 3.1). Heavy biofilm accumulation may require flossing and a toothbrush prophylaxis to aid in the examination process. Occasionally a gross debridement must be schedule before final clinical examination of the teeth may be accomplished.

Clinical Examination for Caries

Contemporary caries management, which encompasses expanded nonoperative approaches and conservative operative interventions,



• **Fig. 3.1** Accurate clinical examination requires a clean, dry, well-illuminated mouth. Cotton rolls are placed in the vestibular space and under the tongue to maintain dryness and enhance visibility.

relies on enhanced risk assessment and improved lesion detection and classification. Clinical caries lesion detection has been found lacking; thus improvement is needed.[2] One means of addressing these concerns has been the development of a visual system for caries lesion detection and classification. The International Caries Detection and Assessment System (ICDAS) was developed to serve as a guide for standardized visual caries assessment that could be used for clinical practice, clinical research, education, and epidemiology (Fig. 3.2). In the United States, the Caries Management by Risk Assessment (CAMBRA) movement, as discussed in Chapter 2, embraces the principles of the ICDAS for visual examination and assessment of caries lesions. The ICDAS has been further condensed by the American Dental Association into the Caries Classification System (ADA CCS)[3] (see Fig. 3.2). Assessment of dental caries also requires identification of caries lesion activity so as to make decisions relative to treatment recommendations (Table 3.1). The objective of improved detection and classification systems is to accurately identify those early enamel lesions that are most likely to be reversed by remineralization. Therefore appropriate nonoperative care may be attempted, and lesions that require operative treatment may be identified as early as possible in the disease process. With this approach, restoration

ADA CCS		Initial		Moderate		Extensive	
Occlusal Protocol ***							
ICDAS code	0	1	2	3	4	5	6
							
Definitions	Sound tooth surface; no caries change after air drying (5 sec); or hypoplasia, wear, erosion, and other noncaries phenomena	First visual change in enamel; seen only after air drying or colored, change "thin" limited to the confines of the pit and fissure area	Distinct visual change in enamel; seen when wet, white or colored, "wider" than the fissure/fossa	Localized enamel breakdown with no visible dentin or underlying shadow; discontinuity of surface enamel, widening of fissure	Underlying dark shadow from dentin, with or without localized enamel breakdown	Distinct cavity with visible dentin; frank cavitation involving less than half of a tooth surface	Extensive distinct cavity with dentin; cavity is deep and wide involving more than half of the tooth
Histologic depth		Lesion depth in P/F was 90% in the outer enamel with only 10% into dentin	Lesion depth in P/F was 50% inner enamel and 50% into the outer 1/3 dentin	Lesion depth in P/F with 77% in dentin	Lesion depth in P/F with 88% into dentin	Lesion depth in P/F with 100% in dentin	Lesion depth in P/F 100% reaching inner 1/3 dentin
Sealant/restoration Recommendation for low risk	Sealant optional DIAGNOdent may be helpful	Sealant optional DIAGNOdent may be helpful	Sealant optional or caries biopsy if DIAGNOdent is 20-30	Sealant or minimally invasive restoration needed	Minimally invasive restoration	Minimally invasive restoration	Minimally invasive restoration
Sealant/restoration Recommendation for moderate risk	Sealant optional DIAGNOdent may be helpful	Sealant recommended DIAGNOdent may be helpful	Sealant optional or caries biopsy if DIAGNOdent is 20-30	Sealant or minimally invasive restoration needed	Minimally invasive restoration	Minimally invasive restoration	Minimally invasive restoration
Sealant/restoration Recommendation for high risk * and extreme risk **	Sealant recommended DIAGNOdent may be helpful	Sealant recommended DIAGNOdent may be helpful	Sealant optional or caries biopsy if DIAGNOdent is 20-30	Sealant or minimally invasive restoration needed	Minimally invasive restoration	Minimally invasive restoration	Minimally invasive restoration

* Patients with one (or more) cavitated lesion(s) are high-risk patients. ** Patients with one (or more) cavitated lesion(s) and xerostomia are extreme-risk patients.

*** All sealants and restorations to be done with a minimally invasive philosophy in mind. Sealants are defined as confined to enamel. Restoration is defined as in dentin. A two-surface restoration is defined as a preparation that has one part of the preparation in dentin and the preparation extends to a second surface (note: the second surface does not have to be in dentin). A sealant can be either resin-based or glass ionomer. Resin-based sealants should have the most conservatively prepared fissures for proper bonding. Glass ionomer should be considered where the enamel is immature, or where fissure preparation is not desired, or where rubber dam isolation is not possible. Patients should be given a choice in material selection.

• **Fig. 3.2** American Dental Association Caries Classification System (ADA CCS) and International Caries Detection and Assessment System (ICDAS) chart showing visual caries detection. (Modified from Young DA, Nový BB, Zeller GG, et al.: The American Dental Association Caries Classification System for Clinical Practice, A report of the American Dental Association Council on Scientific Affairs, *J Am Dent Assoc* 146(2):79–86, 2015; and Jenson L, Budenz AW, Featherstone JD, et al.: Clinical protocols for caries management by risk assessment, *J Calif Dent Assoc* 35:714, 2007.)

TABLE 3.1 Characteristics of Active and Inactive Caries Lesions

	CARIES LESION ACTIVITY ASSESSMENT DESCRIPTORS	
Activity Assessment Factor	**Likely to Be Inactive/Arrested**	**Likely to Be Active**
Location of the Lesion	Lesion is not in a plaque stagnation area	Lesion is in a plaque stagnation area (pit/fissure, approximal, gingival)
Plaque Over the Lesion	Not thick or sticky	Thick and/or sticky
Surface Appearance	Shiny; color: brown-black	Matte/opaque/loss of luster; color: white-yellow
Tactile Feeling	Smooth, hard enamel/hard dentin	Rough enamel/soft dentin
Gingival Status (If the Lesion is Located Near the Gingiva)	No inflammation, no bleeding on probing	Inflammation, bleeding on probing

From Young DA, Nový BB, Zeller GG, et al.: The American Dental Association Caries Classification System for Clinical Practice, A report of the American Dental Association Council on Scientific Affairs, *J Am Dent Assoc* 146(2):79–86, 2015.

will result in the removal of the minimum amount of tooth structure.

Caries lesions may be detected by visual changes in tooth surface texture or color or in tactile sensation when an explorer is used judiciously to detect surface roughness by gently stroking across the tooth surface. Current thinking finds that the use of an explorer in this manner might have some relevance for assessing caries activity. However, it cannot be overemphasized that the *explorer must not be used to determine a "stick"* (i.e., a resistance to withdrawal from a fissure or pit). This improper use of a sharp explorer has been shown to irreversibly damage the tooth by turning a sound, remineralizable subsurface lesion into a possible cavitation that is prone to progression. Forcing an explorer into pits and fissures also theoretically risks cross-contamination from one probing site to another. In contrast, for assessment of root caries, an explorer is valuable for detecting root surface softness. Additional methods used in caries lesion identification include radiographs, which show changes in tooth density from normal, and adjunctive tests that use various technologies to aid in caries lesion detection and caries activity (discussed in later sections).

Occlusal Surfaces

Caries lesions are most prevalent in the faulty pits and fissures of the occlusal surfaces where the developmental enamel lobes of posterior teeth partially or completely failed to coalesce (Fig. 3.3A). It is important to remember the distinction between primary occlusal grooves and fossae and occlusal fissures and pits. Primary occlusal grooves and fossae are smooth "valley or saucer" landmarks that result from complete coalescence of developmental enamel lobes (see Chapter 1). Normally, such grooves and fossae are not susceptible to caries because they are not niches for biofilm and frequently are cleansed by the rubbing action of food during mastication. Conversely, occlusal fissures and pits are deep, tight crevices or holes in enamel, where the lobes failed to coalesce partially or completely. Fissures and pits are detected visually and may frequently be stained but not diseased.

As noted earlier, sharp explorers previously have been used to evaluate fissures and pits in an attempt to diagnose fissure/pit caries. However, numerous studies have found that the use of a sharp explorer for this purpose did not increase diagnostic validity compared with visual inspection alone.[4-7] The use of the dental explorer for this purpose was found to fracture enamel and serve as a source for transferring pathogenic bacteria among various teeth.[8,9] Therefore the use of a sharp explorer in diagnosing pit-and-fissure caries is contraindicated as part of the detection process.

An occlusal surface is examined visually and radiographically.[10,11] The visual examination is conducted in a dry, well-illuminated field. Direct vision is used to observe how light passes into the surface of the tooth structure. The occlusal surface is diagnosed as diseased if external chalkiness (enamel caries) or subsurface opacity (dentin caries) or cavitation of tooth structure, forming the fissure or pit, is seen. At times a brown-gray discoloration, radiating peripherally from the fissure or pit, is present (see Fig. 3.3A, enamel area adjacent to the central pit/lingual fissure) indicating caries progression in dentin below the translucent enamel. In contrast, it is common to observe nondiseased occlusal surfaces with narrow grooves or fossae that exhibit superficial staining, but no visual changes in light reflection through the enamel immediately adjacent (see Fig. 3.3A, distal aspect of central groove and distal fossa area) and with no radiographic evidence of caries. The superficial staining is extrinsic and occurs over several years of oral exposure in a person with low caries risk. Caries lesions occasionally develop on cusp tips (see Fig. 3.3B). Typically, these are the result of developmental enamel defects or following loss of enamel (exposure of dentin) due to erosion, abrasion, or parafunction. The presence of caries in these self-cleansing areas usually indicates that the patient is at high risk of developing additional caries (see Risk Assessments and Profiles in the Diagnosis section to come). Carious pits and fissures also occur on the occlusal two thirds of the facial or lingual surface of posterior teeth and on the lingual surface of maxillary incisors.

The clinical interpretation of subtle changes in the appearance of tooth structure is aided by simultaneous consideration of the patient's overall caries risk, along with the patient's previous patterns of susceptibility. The patient's medical history, dental history, oral hygiene, diet, and age, among other caries risk factors and indicators, may suggest a prediction of current and future caries activity. In addition, occlusal caries lesions tend to occur bilaterally.

The ICDAS uses a two-stage process to record the status of the caries lesion. The first is a code for the severity of the caries lesion and the second is for the restorative status of the tooth. The status of the caries severity is determined visually on a scale of 0 to 6:

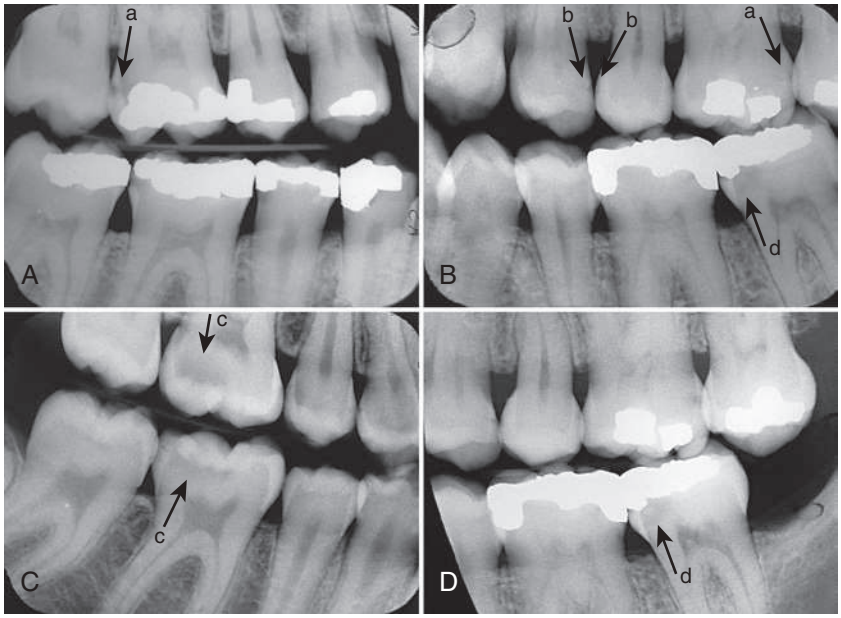
0 = sound tooth structure
1 = first visual change in enamel
2 = distinct visual change in enamel
3 = enamel breakdown, no dentin visible
4 = dentinal shadow (not cavitated into dentin)
5 = distinct cavity with visible dentin
6 = extensive distinct cavity with visible dentin



• **Fig. 3.3** Caries may be diagnosed clinically by careful inspection. A, Loss of translucency and change in color of occlusal enamel resulting from a carious fissure. B, Caries lesions on cusp tips. C, White chalky appearance or shadow under marginal ridge (distal #4 and mesial #5). D, Incipient smooth-surface caries lesion, or a white spot, has intact surface. E, Smooth-surface caries may appear white or dark, depending on the degree of extrinsic staining. F, Root-surface caries.

This severity code is paired with a restorative/sealant code 0 to 8:

0 = not sealed or restored
2 = sealant, partial
3 = sealant, full; tooth-colored restoration
4 = amalgam restoration
5 = stainless steel restoration
6 = ceramic, gold, PFM (porcelain-fused-to-metal) crown or veneer
7 = lost or broken restoration
8 = temporary restoration



• **Fig. 3.4** Caries may be diagnosed radiographically as translucencies in the enamel or dentin. A and B, Proximal caries tends to occur bilaterally (*a*) and on adjacent surfaces (*b*). C, Occlusal caries (*c*). D, Recurrent caries gingival to an existing restoration (*d*). This same recurrent caries (*d*) also is shown in B. Note that there are additional radiolucencies (consistent with caries lesion development) that are not identified with arrows.

See Fig. 3.2 for examples of ADA CCS and ICDAS coding for caries lesion severity. The details of the ICDAS system for detection, and training to use the system with an online tutorial, are available at www.icdas.org.

Proximal Surfaces

Early proximal surface caries, one form of smooth-surface caries, is usually diagnosed radiographically (Fig. 3.4A and B). It also may be detected by careful visual examination after tooth separation or through fiberoptic transillumination.[12] When the caries lesion has progressed through the proximal surface enamel and has demineralized dentin, a white opaque appearance or a shadow under the marginal ridge may become evident (see Fig. 3.3C). Careful probing with an explorer on the proximal surface may detect cavitation, which is defined as a break in the surface contour of enamel. The combined use of all examination methods may be helpful in arriving at an accurate final diagnosis.

Brown spots on intact, hard proximal surface enamel adjacent to and usually gingival to the contact area are often seen in older patients, in whom caries activity is low. These discolored areas are a result of extrinsic staining during earlier caries demineralizing episodes, each followed by a remineralization episode. These areas are no longer carious and are usually more resistant to caries as a result of fluorohydroxyapatite formation. Restorative treatment of these areas is not indicated. Inactive proximal caries lesions sometimes are difficult to correctly diagnose because of faint radiographic evidence revealing previous mineral loss.

Proximal surface caries in anterior teeth may be identified by radiographic examination, visual inspection (with optional transillumination), or probing with an explorer. Transillumination is accomplished by placing the mirror or light source on the lingual aspect of teeth and directing the light through teeth. Small early enamel proximal lesions may be detectable only on the radiograph (see Fig. 3.4B). More advanced proximal lesions appear as a dark area along the marginal ridge when the light is directed through the tooth. In addition to transillumination, tactile exploration of anterior teeth is appropriate to detect cavitation because the proximal surfaces generally are more visible and accessible than in the posterior regions.

Another form of smooth-surface caries may occur on the facial and lingual surfaces of the teeth of patients with high caries activity, particularly in the cervical areas that are less accessible for cleaning. The earliest clinical evidence of early enamel lesions on these surfaces is a white spot that is visually different from the adjacent translucent enamel that appears when the surface is dried. Rewetting results in partial or total disappearance. This appearing–disappearing phenomenon distinguishes the smooth-surface early enamel lesion from the enamel white spot that results from *nonhereditary enamel hypocalcification* (see section on clinical examination for additional defects). Both types of white spots are undetectable tactilely because the surface is intact, smooth, and hard. For white spot lesions, nonsurgical remineralization therapies (discussed in Chapter 2) should be instituted to promote remineralization.

The presence of several facial (or lingual) smooth-surface caries lesions within a patient's dentition suggests a high caries rate, which means that if the existing risk factors are not addressed, the patient is at high risk for developing more lesions in the future. In a caries-susceptible patient, the gingival third of the facial surfaces of maxillary posterior teeth and the gingival third of the facial and lingual surfaces of mandibular posterior teeth should be evaluated carefully because these surfaces are often at a greater risk for caries. Advanced smooth-surface caries exhibits discoloration and demineralization and feels soft as the explorer is translated across the suspicious area. The discoloration may range from white to dark brown, with rapidly progressing caries usually being light in color. Slowly progressing caries, in a patient with low caries activity, darkens over time because of extrinsic staining and physical changes in the structure of the dentin collagen matrix. Remineralization

of the decalcified tooth structure will return the tactile hardness of the lesion and is an evidence that the caries has been arrested. The dentin in an arrested remineralized lesion has become sclerotic (see Chapter 1). Such an arrested lesion at times may be rough, although cleanable, and restoration is not indicated except to address the esthetic concerns of the patient or to assist with patient control of biofilm accumulation. These lesions are inactive but remain susceptible to new caries activity in the future.

Cervical Areas

In patients with attachment loss, extra care must be taken to inspect for root-surface caries. A combination of root exposure, dietary changes, systemic diseases, and medications that affect the amount and character of saliva may predispose a patient, especially an older individual, to root-surface caries. Lesions are often found at the cementoenamel junction (CEJ) or more apically on cementum or exposed dentin in older patients or in patients who have undergone periodontal surgery (see Fig. 3.3F). Early in its development, root caries appears as a well-defined, discolored area adjacent to the gingival margin, typically near the CEJ. Root caries is softer than the adjacent tooth structure, and lesions typically spread laterally around the CEJ. Although no clinical criteria are universally accepted for the diagnosis of root caries, it is generally agreed that softened cemental or dentinal tooth structure compared with the surrounding surface is characteristic.[13] Active root caries is detected by the presence of softening and cavitation.[14,15] Although root-surface caries may be detected on radiographic examination, a careful, thorough clinical examination is crucial. A difficult diagnostic challenge is the patient who has attachment loss with no gingival recession, limiting accessibility for clinical inspection of the proximal root surfaces. Proximal root-surface lesions often progress rapidly and are best diagnosed using quality bitewing radiographs. Differentiation of a caries lesion from a radiolucent artifact created by radiographic cervical burnout is, however, essential.[12,13]

Clinical Examination of Amalgam Restorations

Evaluation of existing restorations should be accomplished systematically in a clean, dry, well-lit field. Clinical evaluation of amalgam restorations requires visual observation, application of tactile sense with the explorer, use of dental floss, interpretation of radiographs, and knowledge of the probabilities that a given condition is sound or at risk for further breakdown. At least 11 distinct conditions might be encountered when amalgam restorations are evaluated: (1) amalgam "blues," (2) proximal overhangs, (3) marginal ditching, (4) voids, (5) fracture lines, (6) lines indicating the interface between abutted amalgam restorations placed at separate times, (7) improper anatomic contours, (8) marginal ridge incompatibility, (9) improper proximal contacts, (10) improper occlusal contacts, and (11) recurrent caries lesions.

Discolored areas or "amalgam blues" are often seen through the enamel in teeth that have amalgam restorations. This bluish hue results either from the leaching of amalgam corrosion products into the dentinal tubules or from the color of underlying amalgam seen through translucent enamel. The latter occurs when the enamel has little or no dentin support, such as in undermined cusps, marginal ridges, and regions adjacent to proximal margins. When other aspects of the restoration are sound, amalgam blues do not indicate caries, do not warrant classifying the restoration as defective, and require no further treatment. Replacement of the restoration may be considered, however, for elective improvement of esthetics or for areas under heavy functional stress that may require a cusp coverage restoration designed to prevent possible tooth fracture.



• **Fig. 3.5** Proximal restoration overhang (*a*) may be diagnosed radiographically.

Proximal overhangs are diagnosed visually, tactilely, and radiographically (Fig. 3.5). The amalgam–tooth junction is evaluated by moving the explorer back and forth across it. If the explorer stops at the junction and then moves outwardly onto the amalgam, an overhang is present. Overhangs also may be confirmed by the catching or tearing of unwaxed dental floss. Such an overhang likely represents an area of biofilm accumulation, provides an obstacle to good oral hygiene, and may contribute to chronic inflammation of adjacent soft tissue. This type of overhang should be corrected, and often indicates the need for restoration replacement.

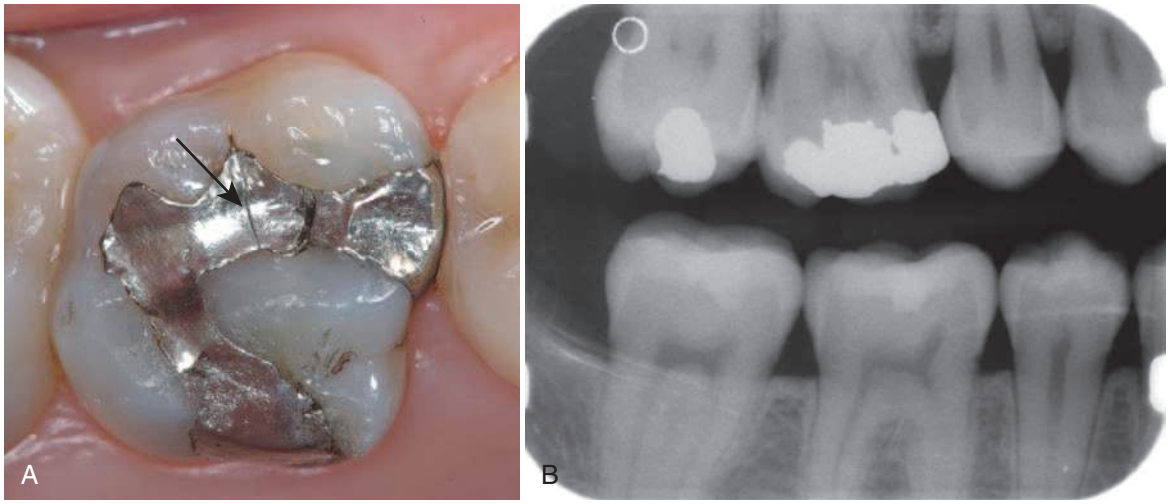
Marginal gap formation (or "ditching") is the deterioration of the amalgam–tooth interface as a result of enamel wear and/or restoration edge fracture (Fig. 3.6A). Improper tooth preparation may predispose an amalgam restoration to ditching. It can be diagnosed visually or by the explorer dropping into an opening as it crosses the margin. Shallow ditching less than 0.5 mm deep usually is not a reason for restoration replacement because the area is self-cleaning and not prone to caries development.[16] Such a restoration usually looks worse than it really is. The ongoing self-sealing property of amalgam allows the restoration to continue serving adequately if it can be satisfactorily cleaned and maintained. If the ditch is too deep to be cleaned or jeopardizes the integrity of the remaining restoration or tooth structure, the restoration should be replaced.[16] However, marginal gaps near the gingival wall frequently become areas of secondary caries development and correction of these areas is indicated.[17]

Localized voids, which result from poor condensation of the amalgam, may also occur at the margins of amalgam restorations. If the void is at least 0.3 mm deep and is located in the gingival third of the tooth crown, the restoration is judged as defective and should be repaired or replaced. Accessible small voids in other marginal areas where the enamel is thicker may be corrected by enamel recontouring or repairing with a small restoration.

A careful clinical examination is able to detect the presence of a fracture line across the occlusal portion of an amalgam restoration. A line that occurs in the isthmus region generally indicates a fractured amalgam, and the defective restoration must be replaced (Fig. 3.7A). Procedures involved with replacement must ensure adequate thickness of the amalgam restoration and rounding of the internal line angles (e.g., the axiopulpal line angle) so as to limit the likelihood of recurrence of a fracture on the occlusal surface (see Fig. 3.7B). Care must be taken to correctly evaluate any such line, however, especially if it is in the midocclusal area because this may be an interface line, a manifestation of two abutted restorations accomplished at separate appointments (see Fig. 3.7A). If other aspects of the abutted restorations are satisfactory, replacement is unnecessary.



• **Fig. 3.6** Restorations may be diagnosed clinically as being defective by observing the following. A, Deep marginal ditching. B, Improper contour. C, Recurrent caries. D, Esthetically unappealing dark staining.



• **Fig. 3.7** Lines across the occlusal surface of an amalgam restoration. A, Fracture line indicates replacement. An interface line *(arrow)* indicates two restorations placed at separate appointments, which, by itself, is insufficient indication for replacement. B, Radiograph revealing thin amalgam area, which allowed material flexure and subsequent fracture.

Amalgam restorations should duplicate the normal anatomic contours of teeth. Restorations that impinge on soft tissue, have inadequate embrasure form or proximal contact, or prevent the use of dental floss should be classified as defective, indicating recontouring or replacement (see Fig. 3.6B).

The marginal ridge portion of the amalgam restoration should be compatible with the adjacent marginal ridge. Both ridges should be at approximately the same level and display correct occlusal embrasure form for passage of food to the facial and lingual surfaces and for proper proximal contact area (see Chapter 1). If the marginal ridges are incompatible and are associated with poor tissue health, food impaction, or the inability of the patient to floss, the restoration is defective and should be recontoured or replaced.

The proximal surface of an amalgam restoration should recreate the normal height of contour such that it comes into contact with the adjacent tooth at the proper occlusogingival and faciolingual area with correct adjacent embrasure form (a "closed" contact). The use of floss is helpful in assessing the intensity of a closed contact. If the proximal contact of any restoration is suspected to be inadequate, it should be evaluated visually by trial angulations of a mouth mirror (held lingually when viewing from the facial aspect, etc.) to reflect light and see if a space where the contact should occur ("open" contact) is present. For this viewing, the contact must be free of saliva. If the contact is open and is associated with poor interproximal tissue health, food impaction, or both, the restoration should be classified as defective and should be replaced or repaired. An open contact typically is annoying to the patient, so correcting the problem usually is an appreciated service.

Recurrent caries adjacent to the marginal area of the restoration is detected visually, tactilely, or radiographically and is an indication for repair or replacement (see Figs. 3.4D and 3.6C). The same criteria for initial proximal and occlusal caries lesions apply for the diagnosis of and intervention for recurrent caries lesions around restorations.

Improper occlusal contacts on an amalgam restoration may cause deleterious occlusal loading (and predisposition to fracture or pain on biting from hyperocclusion), undesirable tooth movement, or both. Premature occlusal contacts may be seen as a "shiny" spot on the surface of the restoration or detected by occlusal marking paper. Such a condition warrants correction by selective occlusal adjustment.

Clinical Examination of Indirect Metal Restorations

Indirect metal restorations should be evaluated clinically in the same manner as amalgam restorations. Any aspect of the restoration that is not satisfactory, that is causing harm to tissue or occlusal function, should be noted and considered for recontouring, repair, or replacement.

Clinical Examination of Composite and Other Tooth-Colored Restorations

Tooth-colored restorations (direct and indirect) should be evaluated clinically in the same manner as amalgam and cast-metal restorations. The presence of improper contour or inadequate proximal contact, overhanging margin, recurrent caries, or occlusal interference should be noted and considered for correction. Corrective procedures include recontouring, polishing, repairing, or replacement of the restoration.

One of the main concerns with anterior teeth is esthetics. If a tooth-colored restoration has dark marginal staining or is discolored to the extent that it is esthetically unappealing to the patient, the restoration should be judged as defective (see Fig. 3.6D). Marginal staining that is judged to be noncarious may be corrected by a small repair restoration along the margin. Occasionally, the staining is superficial and may be removed by resurfacing or removal of restoration excess extending beyond the preparation margins.

Clinical Examination of Dental Implants and Implant-Supported Restorations

Baseline radiographs that reveal the initial levels of implant bone support should be obtained when the implant is restored. Percussion of the restoration should reveal a clinical sound consistent with integration. Probing depths associated with the implant fixture should be consistent with the thickness of the local gingival tissue. The gingival tissue should be assessed for signs of inflammation (redness, edema, tenderness, bleeding on probing). The marginal adaptation between implant restorations and their abutments should allow for optimal biofilm removal. Any deviation from normal should be noted. Many edentulous areas receive implants that are generally smaller than the roots of the teeth they are replacing. Therefore the restorations of the implants require modified cervical contours. Implant restorations should be evaluated for proper cervical (especially proximal) contours that limit food impaction or biofilm accumulation.

Chronic inflammation (periimplantitis), secondary to the presence of residual dental cement or biofilm accumulation, of the tissue immediately adjacent to the implant fixture/restoration may lead to localized bone loss around an implant and impact its long-term survival. Periimplantitis has a multifactorial etiology. See Chapter 11, and a textbook dedicated to dental implantology, for additional information.

Clinical Examination for Additional Defects

A thorough clinical examination occasionally identifies localized noncavitated, hard white areas on the facial (Fig. 3.8) or lingual surfaces or on the cusp tips of teeth. Generally, these are hypocalcified areas of enamel resulting from childhood fever, trauma, or fluorosis that occurred during the developmental stages of tooth formation. These areas are diagnosed as *nonhereditary developmental enamel hypoplasia*. Another cause of hypocalcification is arrested and remineralized incipient caries, which leaves an opaque, discolored, and hard surface. When smooth and cleanable, such areas do not warrant restorative intervention unless they are esthetically offensive to the patient. These areas remain visible whether the tooth is wet or dry, and should not be confused with the opaque white smooth-surface incipient caries lesions that appear when



• **Fig. 3.8** Nonhereditary hypocalcified areas on facial surfaces. These areas may result from numerous factors but do not warrant restorative intervention unless they are esthetically offensive or cavitation is present.

teeth are air-dried. Care must be exercised in distinguishing nonhereditary developmental enamel hypoplasia from an early enamel caries lesion.

Rare genetic disorders affecting enamel and dentin may be discovered during clinical examination. Defective enamel organization and calcification, which results in teeth that are compromised in appearance and strength, is referred to as *amelogenesis imperfecta*. Defective dentin formation and a compromised dentinoenamel junction (DEJ) resulting in early loss of clinically normal enamel is referred to as *dentinogenesis imperfecta*. Additional information on these genetic disorders may be found in textbooks on oral pathology.

The loss of surface tooth structure by chemical action in the continued presence of demineralizing agents with low pH (Fig. 3.9) is defined as *erosion*. The resulting defective surface is usually smooth. Although erosive agents are the predominant causative factors, it is thought that toothbrushing and/or other abrasive agents in the diet may accelerate the loss of tooth structure, which is generally referred to as *erosive tooth wear*. It is necessary to document the severity of the tooth structure loss and the specific areas that have been affected. If the defects are only on the lingual of upper teeth, the diagnosis would be different from finding defects on the occlusal surfaces of lower molars. Exogenous acidic agents such as lemon juice (through sucking on lemons) may cause crescent-shaped or dished defects (rounded as opposed to angular) on the surfaces of teeth exposed to the agent (see Fig. 3.9A), whereas endogenous acidic agents, such as gastric fluids, cause generalized erosion on the lingual, incisal, and occlusal surfaces (see Fig. 3.9B). Erosion processes may also be involved in the loss of the tooth structure with a clinical presentation of "cupped-out" areas on occlusal surfaces. These defective areas are associated with the binge–purge syndrome in bulimia, or with gastroesophageal reflux disease (GERD). Many patients with GERD are often not aware of their gastric symptoms or do not associate them with the problems with their teeth. Consultation with a physician to obtain a proper diagnosis of GERD may assist in the diagnosis and



• **Fig. 3.9** Erosion. A, Crescent-shaped defects on enamel facial surfaces caused by exogenous demineralizing agent (from sucking on lemons several years previous to the time of the photograph). B, Generalized erosion caused by endogenous fluids. C, Rounded cervical lesions associated with toothbrush abrasion. D, Idiopathic erosion lesions in cervical areas are hypothesized to be associated with abnormal occlusal forces. E, Generalized attrition caused by excessive functional or parafunctional mandibular movements. F, Enamel craze lines.

management of erosion. Other sources of erosion may be use of sports drinks, herbal teas, and vomiting associated with chemotherapy, and, in the case of alcoholism, the presence of stomach contents in the mouth during periods of excessive alcohol consumption. It is necessary to document the erosion process as it progresses over time through the use accurate study models, photography, and/or digital scanning technology. Risk assessments for erosion should be included in the assessment of the patient, as indicated. The flow and buffering capacity of the individual patient's saliva impact the rate of progression of erosive tooth wear. Understanding, identifying, and diagnosing tooth damage secondary to erosion (erosive tooth wear) is essential if management of the disease process is to be successful.[18]

Abnormal tooth surface loss resulting from direct frictional forces between teeth and external objects or from frictional forces between contacting teeth in the presence of an abrasive medium is termed *abrasion.* Habitual chewing on hard objects (e.g., paper clips, pens, pencils) or chronic use of agents with high abrasivity (e.g., smokeless tobacco, inadequately washed vegetables) may result in loss of occlusal-surface tooth structure.

The loss of tooth structure in the cervical areas (*abrasion*) is commonly seen as a rounded notch in the gingival portion of the facial aspects of teeth (see Fig. 3.9C). In contrast to cervical lesions that develop from abrasion processes, *idiopathic erosion lesions* ("*abfractions*") are cervical, wedge-shaped defects (angular as opposed to rounded) similar to the defects customarily associated with abrasion but in which one of the possible causative factors may include excessive flexure of the tooth as a result of heavy, eccentric occlusal forces (see Fig. 3.9D). The heavy occlusal loading may also lead to the development of a pronounced occlusal wear facet. It is hypothesized that the flexural force produces tension stress in the affected wedge-shaped region on the tooth side away from the tooth-bending direction, resulting in loss of the surface tooth structure by microfractures, which is termed *abfracture.*[19] Proponents of this hypothesis add that microfractures may increase the rate of tooth structure loss during abrasion from tooth brushing and/or from acids in the diet or biofilm. Opponents of this hypothesis note that these cervical lesions have been detected in individuals who do not have any apparent evidence of heavy occlusal forces (such as wear facets and/or fremitus). The general consensus among experts is that the etiology of these lesions is multifactorial and that well-designed clinical research studies are required to gain better understanding relative to the etiology of cervical abrasion and abfraction defects, which are generally referred to as noncarious cervical lesions (NCCLs).[20] The presence of such defects does not automatically warrant intervention.

The mechanical wear of the incisal or occlusal tooth structure that results from functional or parafunctional movements of the mandible is termed *attrition*. Although a certain degree of attrition is expected with age, it is important to note abnormally advanced attrition (see Fig. 3.9E). If abnormal attrition is present, the patient's functional movements should be evaluated and inquiry made with regard to potential parafunctional habits such as tooth grinding or clenching/grinding *(bruxism)*. Heavy occlusal loading from clenching may result in the presence of "craze lines" that are limited to enamel (i.e., do not progress through the DEJ into dentin; see Fig. 3.9F). Craze lines are not sensitive and do not require treatment but may be evidence of excessive masticatory muscle activity (see Chapter 1). The etiology of the parafunction may include stress, airway issues, and/or sleep apnea. In some older patients, the enamel of the cusp tips (or incisal edges) is worn off, resulting in cupped-out areas because the exposed, softer dentin wears faster than the surrounding enamel, though there may also be an erosive component to the process. Sometimes, these areas are an annoyance because of food retention or the presence of peripheral, ragged, sharp enamel edges. The sharp edges may result in tongue or cheek biting; rounding these edges does not completely resolve the problem but may improve comfort. Slowing such wear by appropriate restorative treatment may be indicated.

The examination process may reveal areas of horizontal or vertical fracture development. Awareness of extreme variations in dental anatomy aids in the identification of fracture-prone areas. For example, deep developmental fissures that cross between marginal or cusp ridges may predispose posterior teeth to fracture. Early fractures may be invisible upon initial assessment. Appropriate dye materials or transillumination may aid in detecting the line of fracture within the tooth structure. Cusp isolation/loading devices and techniques must be utilized so as to identify fractures that involve the dentin and are symptomatic (i.e., fractures that are actively propagating). Teeth with active, symptomatic fractures should be considered for full coverage of the occlusal surface. Fractures that have been present for a period of time become stained and thereby visible during examination; they should be considered at risk for further propagation into dentin. Any tooth that has extensive caries, or restoration, and remaining cusps with little dentin support should be identified as being susceptible to future fracture and considered for a cusp-protecting restoration (Fig. 3.10). Complete cusp fracture is a common occurrence in posterior teeth. In general, the most frequently fractured cusps are the nonfunctional cusps (see Chapter 1). Specifically, the most frequently fractured teeth are mandibular molars and second premolars, with the lingual (nonfunctional) cusps fracturing more often than the facial (functional) cusps. Maxillary premolars also frequently fracture, and similar to mandibular teeth, the facial (nonfunctional) cusps fracture more often than the lingual (functional) cusps. The mesiofacial (nonfunctional) and distolingual (small functional) cusps are the most commonly fractured cusps in maxillary molars.[21] A study of fracture severity found that 95% of the fractures



• **Fig. 3.10** Extensively restored teeth with weakened and fractured cusps. Note the distal developmental fissure in the second molar, which further predisposes the distal cusps to fracture.

exposed dentin, 25% were below the CEJ, and 3% resulted in pulp exposure. The consequences of posterior tooth fracture were found to vary, with maxillary premolar and mandibular molar fractures being generally more severe. Most fractures were treated with direct or indirect restorations or recontouring and polishing; 3% of the fractured teeth were extracted, and 4% received endodontic treatment.[22] Risk factors for nontraumatic fracture of posterior teeth were found to be the presence of a stained fracture in enamel and an increase in the proportion of the volume of the natural tooth crown occupied by a restoration.[23,24] The examination process should notate the presence and activity of all fracture areas.

The dental examination also may reveal dental anomalies that include variations in size, shape, structure, or number of teeth—such as dens in dente, macrodontia, microdontia, gemination, concrescence, dilaceration, amelogenesis imperfecta, and dentinogenesis imperfecta. An in-depth discussion of these anomalies is beyond the scope of this text. The reader should consult an oral pathology textbook for additional information.

Radiographic Examination of Teeth and Restorations

Radiographs are an indispensable part of the contemporary dentist's diagnostic armamentarium. The use of diagnostic ionizing radiation is, however, not without risks. Cumulative exposure to ionizing radiation potentially may result in adverse effects. The diagnostic yield or potential benefit that might be gained from a radiograph must be weighed against the financial costs and the potential adverse effects of exposure to radiation. Several technologies, particularly digital radiography, are now available and are designed to enhance diagnostic yield and reduce radiation exposure.

The ADA, in collaboration with the Food and Drug Administration (FDA), developed guidelines for the prescription of dental radiographic examinations to serve as an adjunct to the dentist's professional judgment with regard to the best use of diagnostic imaging. Radiographs help the dental practitioner evaluate and definitively diagnose many oral diseases and conditions. However, the dentist must weigh the benefits of taking dental radiographs against the risk of exposing a patient to ionizing radiation, the effects of which accumulate from multiple sources over time. The dentist, being aware of the patient's health history and vulnerability to oral disease, is in the best position to make this judgment. For this reason, guidelines are intended to serve as a resource for the practitioner and are not intended to be standards of care, requirements, or regulations. The ADA/FDA guidelines help direct the type and frequency of radiographs needed according to patient condition and risk factors (Table 3.2).

Generally, patients at higher risk for caries or periodontal disease should receive more frequent and more extensive radiographic surveys. A systematic review of methods of diagnosing dental caries lesions found that although radiographs were useful in detecting lesions, they do have limitations.[25] For the examination of occlusal surfaces, radiographs had moderate sensitivity and good specificity for diagnosing dentinal lesions; however, for enamel lesions, the sensitivity was poor and the specificity was reduced (see the section on Diagnosis for description of these terms). Studies of the radiographic examination of proximal surfaces found that there was moderate sensitivity and good specificity for the detection of cavitated lesions and low to moderate sensitivity and moderate to high specificity for enamel or dentinal lesions. Before rendering a diagnosis and deciding on treatment, information obtained from radiographs should be confirmed or augmented through clinical examination findings if at all possible.

For diagnosis of proximal surface caries, restoration overhangs, or poorly contoured restorations, posterior bitewing and anterior periapical radiographs are most helpful. When interpreting the radiographic presentation of proximal tooth surfaces, it is necessary to know the normal anatomic picture presented in a radiograph before any abnormalities may be diagnosed. In a radiograph, a proximal caries lesion usually appears as a dark area or a radiolucency in the enamel slightly apical to the contact (see Fig. 3.4A). This radiolucency is typically triangular and has its apex toward the DEJ.

Moderate-to-deep occlusal caries lesions may be seen as a radiolucency extending into dentin (see Fig. 3.4C). Because the specificity of radiographs for detecting dentinal lesions on occlusal surfaces is relatively good at 80% (very few false positives), when a radiolucency is apparent beneath the occlusal enamel surface emanating from the DEJ a diagnosis of caries is appropriate. However, because the sensitivity of radiographs for dentinal lesions on the occlusal surface is rather low (50%), the absence of a radiolucency does not mean that a lesion is not present. In these situations, the clinician should rely more on the results of the visual examination and the findings of any adjunctive tests (discussed later).

Some defective aspects of restorations, including improper contour, overhangs (see Fig. 3.5), and recurrent caries lesions gingival to restorations (see Fig. 3.4D), may also be identified radiographically. The height and integrity of the marginal periodontium may be evaluated using bitewing radiographs. Pulpal abnormalities such as pulp stones and internal resorption may be identified in various radiographs. Periapical radiographs are helpful in identifying changes in the periapical periodontium that are consistent with periapical abscesses, dental granulomas, or cysts. Impacted third molars, supernumerary teeth, and other congenital or acquired abnormalities also may be discovered on periapical radiographic examination. The sensitivity and specificity of dental radiographs vary, however, according to the diagnostic task (e.g., surface of the tooth being examined, proximal versus occlusal; and depth, enamel versus dentin).

Radiographs aid in determining the relationship between the margins of existing or proposed restorations and bone. A biologic width of at least 2 mm is required for the junctional epithelium and the connective tissue attachments located between the base of the sulcus and the alveolar bone crest (Fig. 3.11A). In addition to this physiologic dimension, the restoration margin should be placed occlusally as far away as possible from the base of the sulcus. Restoration margins that encroach on the biologic width, by being placed deep in the sulcus, are inaccessible for biofilm removal and result in chronic inflammation. The inflammatory state may be clinically detected as clinical redness, swelling, and bleeding on probing or flossing in the area. Localized loss of osseous support will occur and the biologic width will reorganize further apically. The osseous loss and reorganization will result in deeper periodontal probing depths, which in turn will further limit effective biofilm removal. For these reasons, the final position of a proposed gingival margin, which is dictated by the existing restoration, caries, or retention features, must be estimated before restoration to determine if crown-lengthening procedures are indicated (see Fig. 3.11B). Surgical crown lengthening procedures involve the surgical removal of the gingiva, bone, or both to create a longer clinical crown and provide more tooth structure for placing the restoration margin and for increasing retention form. Another possible treatment option may be to orthodontically extrude the tooth so that the restoration margins do not violate the biologic width. Restorative

procedures must be accomplished such that the periodontal health may be maintained.

Dental radiographs should always be interpreted cautiously. The dental radiograph is a two-dimensional image of a three-dimensional mass; thus a facial or lingual lesion (or radiolucent tooth-colored restoration) may be radiographically superimposed over the proximal area, mimicking a proximal caries lesion (false positive). The general finding that approximately 25% mineral loss has to occur before a radiolucency begins to appear on a radiograph means that a caries lesion may be present and not detected (false negative). Misdiagnosis may occur when cervical burnout (the radiographic picture of the normal structure and contour of the cervical third of the crown) mimics a caries lesion. Finally, although a caries lesion may be more extensive clinically than it appears radiographically, it is estimated that over half of radiographically detected proximal lesions (in the outer half of dentin) are likely to be noncavitated and treatable with remineralization measures.[26] Radiographic findings must always be clinically verified (if possible) prior to the finalization of a diagnosis and treatment plan.

Adjunctive Aids for Examining Teeth and Restorations

Magnification in Operative Dentistry

Clinical dentistry often requires the viewing and evaluation of small details in teeth, intraoral and perioral tissues, restorations, and study casts. Unaided vision is often inadequate to view details

TABLE 3.2 Guidelines for Prescribing Dental Radiographs

The recommendations in this chart are subject to clinical judgment and may not apply to every patient. They are to be used by dentists only after reviewing the patient's health history and completing a clinical examination. Because every precaution should be taken to minimize radiation exposure, protective thyroid collars and aprons should be used, whenever possible. This practice is strongly recommended for children, women of childbearing age, and pregnant women.

Type of Encounter	Patient Age and Dental Developmental Stage				
	Child With Primary Dentition (Prior to Eruption of First Permanent Tooth)	Child With Transitional Dentition (After Eruption of First Permanent Tooth)	Adolescent With Permanent Dentition (Prior to Eruption of Third Molars)	Adult, Dentate or Partially Edentulous	Adult, Edentulous
New patient* being evaluated for dental diseases and dental development	Individualized radiographic exam consisting of selected periapical/occlusal views and/or posterior bitewings if proximal surfaces cannot be visualized or probed. Patients without evidence of disease and with open proximal contacts may not require a radiographic exam at this time.	Individualized radiographic exam consisting of posterior bitewings with panoramic exam or posterior bitewings and selected periapical images.	Individualized radiographic exam consisting of posterior bitewings with panoramic exam or posterior bitewings and selected periapical images. A full mouth intraoral radiographic exam is preferred when the patient has clinical evidence of generalized dental disease or a history of extensive dental treatment.		Individualized radiographic exam, based on clinical signs and symptoms.
Recall patient* with clinical caries or at increased risk for caries**	Posterior bitewing exam at 6- to 12-month intervals if proximal surfaces cannot be examined visually or with a probe.			Posterior bitewing exam at 6- to 18-month intervals.	Not applicable
Recall patient* with no clinical caries and not at increased risk for caries**	Posterior bitewing exam at 12- to 24-month intervals if proximal surfaces cannot be examined visually or with a probe.		Posterior bitewing exam at 18- to 36-month intervals.	Posterior bitewing exam at 24- to 36-month intervals.	Not applicable
Recall patient* with periodontal disease	Clinical judgment as to the need for and type of radiographic images for the evaluation of periodontal disease. Imaging may consist of, but is not limited to, selected bitewing and/or periapical images of areas where periodontal disease (other than nonspecific gingivitis) can be identified clinically.				Not applicable
Patient for monitoring of growth and development	Clinical judgment as to need for and type of radiographic images for evaluation and/or monitoring of dentofacial growth and development.		Clinical judgment as to need for and type of radiographic images for evaluation and/or monitoring of dentofacial growth and development. Panoramic or periapical exam to assess developing third molars.	Usually not indicated	

TABLE 3.2 Guidelines for Prescribing Dental Radiographs—cont'd

Patient with other circumstances including, but not limited to, proposed or existing implants, pathology, restorative/endodontic needs, treated periodontal disease, and caries remineralization	Clinical judgment as to need for and type of radiographic images for evaluation and/or monitoring in these circumstances.

*Clinical situations for which radiographs may be indicated include but are not limited to:

A. Positive Historical Findings
   1. Previous periodontal or endodontic treatment
   2. History of pain or trauma
   3. Familial history of dental anomalies
   4. Postoperative evaluation of healing
   5. Remineralization monitoring
   6. Presence of implants or evaluation for implant placement

B. Positive Clinical Symptoms/Signs
   1. Clinical evidence of periodontal disease
   2. Large or deep restorations
   3. Deep caries lesions
   4. Malposed or clinically impacted teeth
   5. Swelling
   6. Evidence of dental/facial trauma
   7. Mobility of teeth
   8. Sinus tract ("fistula")
   9. Clinically suspected sinus pathology
   10. Growth abnormalities
   11. Oral involvement in known or suspected systemic disease
   12. Positive neurologic findings in the head and neck
   13. Evidence of foreign objects
   14. Pain and/or dysfunction of the temporomandibular joint and/or muscles of mastication
   15. Facial asymmetry
   16. Abutment teeth for fixed or removable partial prosthesis
   17. Unexplained bleeding
   18. Unexplained sensitivity of teeth
   19. Unusual eruption, spacing, or migration of teeth
   20. Unusual tooth morphology, calcification, or color
   21. Unexplained absence of teeth
   22. Clinical erosion

**Factors increasing risk for caries may include but are not limited to:
   1. High level of caries experience or demineralization
   2. History of recurrent caries
   3. High titers of cariogenic bacteria
   4. Existing restoration(s) of poor quality
   5. Poor oral hygiene
   6. Inadequate fluoride exposure
   7. Prolonged nursing (bottle or breast)
   8. Frequent high sucrose content in diet
   9. Poor family dental health
   10. Developmental or acquired enamel defects
   11. Developmental or acquired disability
   12. Xerostomia
   13. Genetic abnormality of teeth
   14. Many multisurface restorations
   15. Chemotherapy/radiation therapy
   16. Eating disorders
   17. Drug/alcohol abuse
   18. Irregular dental care

From American Dental Association, US Food and Drug Administration: The selection of patients for dental radiograph examinations. Available on www.ada.org. Document created November 2004.

needed to make treatment decisions. Magnification aids such as loupes provide a larger image size for improved visual acuity, while allowing proper upright posture to be maintained with less eye fatigue.

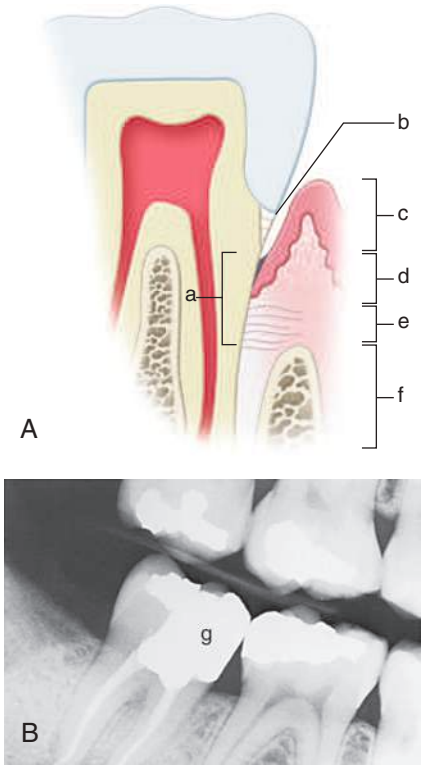
When choosing loupes, several parameters should be considered.[27-29] Magnification (power) describes the increase in image size. Most dentists use magnifications of 2× to 4×. The lower power systems of 2× to 2.5× allow multiple quadrants to be viewed, whereas the higher power systems of 3× to 4× enable viewing of several teeth or a single tooth. In general, higher magnification systems are heavier, have a narrower field of view, are more expensive, and require more light than lower power systems. The use of small, lightweight light-emitting diode (LED) headlamps attached to the eyeglass frame or headband offer the considerable visual advantage of added illumination when used with loupes.

Working distance (focal length) is the distance from the eye to the object when the object is in focus. This parameter should be considered carefully before selecting loupes because the desired working distance depends on the dentist's height, arm length, and seating preferences. Dentists of average height typically choose a working distance of 13 to 14 inches (33–35 cm), whereas tall dentists and those who prefer to work farther away from the patient use working distances of 14 to 16 inches (35–40 cm).

Depth of focus, or the difference between the far and near focus limits of the working distance, depends on the magnification. Typically, the lower the magnification, the greater is the depth of focus.

Many choices of magnification loupes are currently available for dentistry. The simplest magnifiers are the diopter single-lens loupes, which are single-piece plastic pairs of lenses that clip onto eyeglass frames. These loupes are inexpensive and lightweight and may provide magnification of up to 2.5×. However, images may be distorted, and working lengths less than ideal. The more commonly used dental loupe is binocular with lenses mounted on an eyeglass frame. Binocular loupes typically have Galilean and prismatic optics that provide 2×, 3.5×, 4×, and greater magnification. Prescription lenses may be placed in the eyeglass frames for all loupe types. Most models also have side shields or a wraparound design for eye protection and infection control. Two mounting systems are currently available for binocular loupes: (1) flip-up and (2) fixed or through-the-lens types.

Dental microscopes, though limited primarily to endodontic practices in the past, are now being used in some restorative dentistry practices. Compared with high powered loupes, dental microscopes allow the clinician to view intraoral structures at a higher level of magnification while maintaining a broader field of view. Because very small areas may be seen, microscopes are used in detail-oriented procedures such as the finishing of porcelain restoration margins, identifying minute caries lesions, and minimizing the removal of sound tooth structure. Generally, microscopes include five or six levels of magnification that typically range from 2.5× to 20×. Manufacturers of dental microscopes include Carl Zeiss, Inc. (Dublin, CA); Global Surgical Corporation (St. Louis, MO); and Seiler Precision Microscope Instrument Company (St. Louis, MO).



• **Fig. 3.11** A, Biologic width (*a*) is the physiologic dimension needed for the junctional epithelium (*d*) and the connective tissue attachment (*e*), which is measured from the base of the sulcus (*c*) to the level of the bone crest (*f*). The margin of the restoration (*b*) must not violate this dimension. B, Tooth with an existing restoration (*g*) that encroaches on the biologic width requires crown-lengthening procedures before placement of a new restoration.

Cost, size of the equipment, and perceived lack of value to the clinician are factors that have limited the use of microscopes in operative dentistry practice.

Photography in Operative Dentistry

Photography in dentistry has many uses and, with newer digital technologies, is becoming mainstream in dental practice. Photography is an excellent tool for documentation and evaluation. Intraoral cameras and single-lens reflex (SLR) digital cameras provide opportunities to document existing esthetic conditions such as color, shape, and position of teeth. Close-up images of existing pits and fissures provide the opportunity to image current conditions for the purpose of future reevaluation and detection of changes that may be developing. Photographs of preparations of deep caries lesions provide documentation to aid in future diagnosis of tooth conditions. Digital documentation with photographs, and ability to process and store images in an electronic patient record, is easy and cost effective.

Diagnostic Study Models

Study models are helpful in evaluating a patient's clinical status in many situations. Study models are able to provide an understanding of occlusal relationships, help in developing the treatment plan, and serve as a tool for educating the patient. Accurately mounted study models provide an opportunity for a thorough evaluation of the tooth interdigitation, the functional occlusion, and any occlusal abnormalities. Study models allow further evaluation of the plane of occlusion; tilted, rotated, or extruded teeth; crossbites; plunger cusps; wear facets and defective restorations; coronal contours; proximal contacts; and embrasure spaces between teeth. The ability to obtain virtual study models via digital impression technology has increased the ease and level of diagnostic evaluation, especially in situations where the use of conventional impression techniques/materials may not be an option (such as in patients with a hyperactive gag reflex). Combined with clinical and radiographic findings, study casts allow the practitioner to carefully reflect on findings and develop a treatment plan without the patient present, thus saving valuable chair time. When a proposed treatment plan is discussed with the patient, study models are a valuable educational medium in helping the patient understand and visualize existing conditions and the need for the proposed treatment.

Caries Detection Technologies

In addition to the traditional methods of caries detection, several new technologies have emerged and show promising results for the clinical detection and diagnosis of caries lesions. These devices may have the potential to replace the tactile portion of caries detection, where explorers are used to try to estimate the depth of the caries lesions into the pits and fissures. However, these devices have two limitations. The first is that they are only indicated for use on unrestored pits and fissures. The second is that their diagnostic accuracy has not been firmly established. The technologies currently approved by the FDA include laser-induced fluorescence, light-induced fluorescence, and alternating current (AC) impedance spectroscopy (ACIST).[10,30]

The DIAGNOdent device (KaVo Dental Corporation, Charlotte, NC) uses laser fluorescence technology, with the intention of detecting and measuring bacterial products and changes in tooth structure in a caries lesion. This compact and portable device, which requires a clean, dry occlusal surface, yields a numerical score from 0 to 99. The manufacturer has recommended threshold scores that represent the presence and extent of a lesion. A systematic review found that the "device is clearly more sensitive than traditional diagnostic methods, but the increased likelihood of false-positive diagnoses limits its usefulness as a principal diagnostic method."[31]

Another system currently available for caries lesion detection is the CamX Spectra Caries Detection Aid (Air Techniques, Melville, NY). This system claims to detect caries lesions by measuring increased light-induced fluorescence. Special LEDs project high-energy violet or blue light onto the tooth surface. Light of this wavelength supposedly stimulates porphyrins—metabolites unique to cariogenic bacteria—to appear distinctly red, while healthy enamel fluoresces to appear green. Using this fluorescent technology, the data captured by the Spectra system are analyzed by imaging software, which highlights the lesions in different color ranges and defines the potential caries activity on a scale of 0 to 5.

The CarieScan PRO (CarieScan, LLC, Charlotte, NC) is a device for the detection and monitoring of caries by the application and analysis of ACIST (AC Impedance Spectroscopy Technology). The CarieScan PRO claims to enable clinicians to evaluate demineralized tooth structure using ACIST by providing information about tissue being healthy, in the early stages of demineralization, or already significantly decayed. The device provides a color scale and a numerical scale to determine the severity of the caries lesion and is accompanied by management recommendations that range from therapeutic prevention to operative intervention appropriate for the extent of the demineralization.

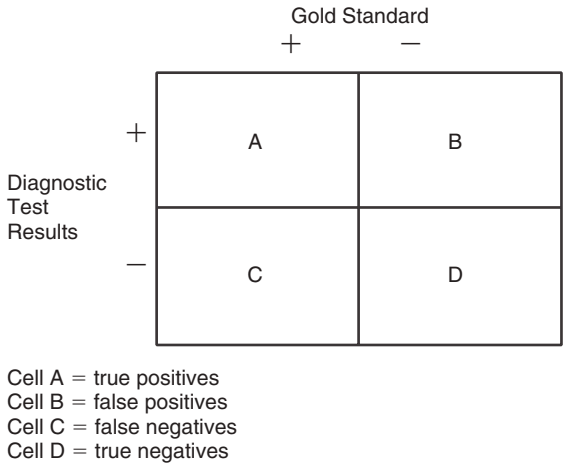
Although these technologies appear promising, the standard of care remains visual inspection of well-illuminated, clean and dry teeth, with use of radiographs as indicated.[32] An ideal diagnostic test accurately detects when a tooth surface is healthy (specificity); when a lesion or demineralization is present (sensitivity); and if demineralization is present, whether or not it is active and whether or not it has cavitated the surface (see section on Diagnosis). Except for the presence of frank cavitation in more advanced lesions, none of the available approaches to detecting caries or determining lesion activity is completely accurate. Thus the clinician must take all of the available diagnostic information together—visual, tactile, radiographic, and so on—along with the respective reported levels of accuracy and combine that with an assessment of the patient's overall caries status to make a final diagnosis of the presence and extent of a caries lesion.

Diagnosis

Dental Disease; Interpretation and Use of Diagnostic Findings

As discussed in Chapter 2, dental caries is a multifactorial, transmissible, infectious oral disease caused primarily by the complex interaction of cariogenic oral flora (biofilm) with fermentable dietary carbohydrates on the tooth surface over time. *Caries lesions are the result of the caries disease process, not the cause.*

The diagnostic effort of health care professionals has been enhanced by the use of principles adopted from clinical epidemiology. This analytic approach relies on "2 × 2" contingency tables (Fig. 3.12) derived from clinical trials data. Such studies compare the results of a diagnostic test with the results obtained from a "gold standard" (knowledge of the actual condition) to determine how well a test identifies the "true," or actual, condition. The results of the diagnostic test, positive or negative, are shown across the rows of the table, and the results of a "gold standard" or the "truth" are displayed in the columns. Cell A of the table contains the cases that the test identifies as being positive (or diseased) that actually are positive (i.e., confirmed by the "gold standard"). These cases are termed *true positives*. Cell B contains all cases for which a positive finding from the diagnostic test is present, but where the actual condition is negative. Therefore this cell denotes *false positives*. Cell C includes the cases identified by the diagnostic test as not being diseased, but actually are diseased, as determined by the "gold standard." Findings in this cell are termed *false negatives*. The final cell, cell D, includes *true negatives*, where the diagnostic test accurately identifies nondiseased cases that are truly negative as confirmed by the "gold standard." A perfect diagnostic test would result in all cases being assigned to cells A or D with no false positives (cell B) or false negatives (cell C).



• **Fig. 3.12** Contingency table for interpretation of diagnostic tests.

When the basics of this table are understood, the information it yields may be put to good use by the diagnostician. The first concept is test *sensitivity*, which is calculated as the number of true positives (A) divided by the number of total positive cases (A + C, i.e., the number of times where disease was actually present regardless of the diagnostic test results). Sensitivity indicates the proportion of individuals with disease in any group or population that is identified positively by the test. In contrast, *specificity* refers to the proportion of individuals without disease properly classified by the diagnostic test and is the ratio of true negatives (D) to all negatives (B + D). Sensitivity and specificity will not vary on the basis of the prevalence of disease, that is, the proportion of cases in a population. Rather, these statistics indicate what proportions of existing disease and absence of disease will be correctly identified in any group of individuals.

A test with low sensitivity indicates that a high probability exists that many of the individuals with negative results have the disease and go undiagnosed. Conversely, a test with high sensitivity means that most of those who actually have disease will be identified as such. Tests with high specificity suggest that patients without the disease are highly likely to test negative. Tests with low specificity will misclassify a sizable proportion as diseased when many are actually free of disease.

Very few tests have *both* high sensitivity and high specificity, so trade-offs are inevitable. The clinician must weigh the seriousness of the disease that is left untreated (in cases of low sensitivity) against the invasiveness of the treatment (in cases of low specificity). In the former, low sensitivity may be acceptable for tests diagnosing slowly progressing, nonfatal conditions but unacceptable for conditions that progress rapidly or are life threatening. In the latter, low specificity may not be acceptable if the treatment is invasive and irreversible, but more acceptable if the treatment is noninvasive and temporary. In the case of dental caries, all things being equal, this means that the clinician may accept a less sensitive test (i.e., miss some initial lesions [cell C]) because caries usually progresses slowly over years. But given that operative treatment is invasive and irreversible, a highly specific test (i.e., few false positives [cell B]) means that fewer healthy teeth will be incorrectly treated.

The dentist should be mindful of the fact that except in cases of relatively large caries lesions, *the accuracy of the methods used to detect lesions (visual inspection, radiographs, caries detection devices, etc.) are all prone to inaccuracies* (Box 3.1). These inaccuracies result in false-positive and false-negative findings. This situation raises the question, "What are the implications of these inaccuracies for clinical decision making?" False-positive findings may result in the surgical treatment of a sound tooth, and false-negative findings will result in a diseased surface receiving remineralization treatment instead of operative treatment. The former situation is irreversible and should be avoided, whenever possible. In the latter situation, false negatives will receive remineralization therapy, regular monitoring, and, if a lesion develops, may be treated operatively at a later time, if needed. This reasonable approach takes into consideration that caries lesions generally do not progress rapidly.[33-35] Thus the clinician should strive to reduce the number of false positives by making sure that strong diagnostic evidence supports the presence

• BOX 3.1 Assessing the Accuracy of a Diagnostic Test for Caries

Contingency Table for Diagnostic Test Evaluation
Histologic Gold Standard
- Caries
- No caries

Diagnostic Test
- Caries
  - True positive (TP)
  - False positive (FP)
- No caries
  - False negative (FN)
  - True negative (TN)

Desirable and Undesirable Outcomes Resulting from Diagnostic Tests with Low Sensitivity or Specificity

Example 1

Diagnosing 100 teeth (90 healthy and 10 carious) with a diagnostic test having a high sensitivity (0.80) and low specificity (0.50) would result in the following:
- Desirable outcomes:
  - Correctly detect 8 of 10 carious teeth (TP)
  - Correctly diagnose 45 of 90 healthy teeth (TN)
- Undesirable outcomes:
  - Fail to detect 2 of 10 carious teeth (FN)
  - Fail to diagnose 45 healthy teeth as carious (FP)

Example 2

Diagnosing 100 teeth (90 healthy and 10 carious) with a diagnostic test having low sensitivity (0.50) and high specificity (0.80) would result in the following:
- Desirable outcomes:
  - Correctly detect 5 of 10 carious teeth (TP)
  - Correctly diagnose 72 of 90 healthy teeth (TN)
- Undesirable outcomes:
  - Fail to detect 5 of 10 carious teeth (FN)
  - Fail to diagnose 18 healthy teeth as carious (FP)

of cavitation or dentin penetration before recommending irreversible operative treatment.

These concepts are widely used in medical practice. Although many of the necessary research studies have not been conducted to develop probabilities for dental conditions, interest in the use of clinical epidemiology in the dental profession has been growing. In the future, more research studies will be conducted so as to provide this information to clinicians, and one should be prepared to take advantage of their use.

Risk Assessments and Profiles

The patient assessment and examination process allows opportunity to identify factors and indicators that increase likelihood (risk) of future problems, given the patient's current behaviors, clinical conditions, and so on (see Chapter 2).[36,37] *Risk assessments* help organize the data relative to multiple causative factors. Few diseases or dental conditions are caused by a single factor. Rather, most diseases and dental conditions have been shown to be associated with numerous behavioral or sociodemographic, physical or environmental, microbiologic, or host factors. In addition, every patient has a different set of risk factors. This presents a challenge to determining the likelihood that a disease or condition would occur in the future or that some form of dental treatment or therapeutics would decrease the chances of disease occurrence. Many risk assessments use terms such as *low risk*, *medium risk*, and *high risk* to associate a level of risk with a category. This is sometimes expressed by using color-coded categories: red for high risk, yellow for medium risk, and green for low risk. Categories simplify the concept for the patient, as they are easily understood while discussing assessments and their implications for treatment recommendations.

Patients who possess risk factors and risk indicators should be considered to be at risk for dental caries even if the examination does not reveal any caries lesions.[33] A patient at high risk for dental caries should receive aggressive intervention to remove or alter as many risk factors as possible. Alternatively, regular monitoring and reassessment might be appropriate for a patient at low risk for dental caries. Risk assessment is a relatively young science in the dental profession, but as more research is completed, evidence is quickly validating this approach to patient care. Approaches to patient care using risk assessments and disease management such as CAMBRA are becoming the recognized standard of care. The CAMBRA guidelines were developed over several years as an evidence-based approach to preventing, reversing, and, when necessary, repairing early damage to teeth caused by caries. Refer to Chapter 2 for more information on how CAMBRA is used to determine caries risk and how this determination helps the clinician in the decision-making process for surgical or nonsurgical therapeutic interventions.

In relation to operative dentistry, risk assessments are made for caries, erosive tooth wear, and structural problems of teeth such as fractures. However, risk assessments should be established for other areas of the stomatognathic system such as periodontal disease, functional occlusal and TMJ issues, and for the "risk" involved in satisfying the patient's esthetic expectations. All treatment for patients should be designed to lower their risk for problems in each of these areas. Dental treatment in any one of the abovementioned areas may improve risk status in that area but at a cost of increased risk in another area. For example, preparation of teeth for full-coverage crowns might reduce occlusal or esthetic risk but at a cost of increasing risk for future caries or pulpal pathology. Taken together, risk assessments provide a *risk profile* that helps guide preventive and operative recommendations that are made to the patient with the goal of mitigating as many risk factors as possible.

Prognosis

Prognosis is the term used to describe the prediction of the probable course and outcome of a disease or condition as well as the outcome expected from an intervention, be it preventive or operative. Prognosis may also be used to estimate the likelihood of recovery from a disease or condition. In operative dentistry, prognosis may be used to describe the likelihood of success of a particular treatment procedure in terms of time of service, functional value, comfort, and esthetic value for the patient. A prognosis may be described as *excellent*, *good*, *fair*, *poor*, or even *hopeless*. Prognosis for a disease or condition is largely dependent on the risk factors and indicators that are present in the patient. However, other variables, such as the skill of the dentist and the current status of the disease before beginning treatment, also have an effect on the prognosis. For example, a patient with severe caries may be willing to eliminate all of the modifiable risk factors, but if the disease is too advanced, the long-term prognosis for the affected teeth may still be poor. It is important for the clinician to take into account the entire risk profile of the patient in all areas of the person's medical and dental health when trying to establish a prognosis. Once the dentist

and the patient have a good understanding of the current condition(s), the patient's risk profile, and all associated prognoses, they will be able to work together as a team to identify treatment options and establish a treatment plan.

Treatment Planning

General Considerations

Patient assessment, examination, and diagnosis result in a listing of dental problems, an inventory of existing risk factors (or indicators), and an accurate prognosis for each tooth and for the patient's overall oral health. The dentist then begins to consider various options in light of the paramount principle in dentistry: to *do no harm*. Clinicians must have a sound knowledge of the current evidence relative to the risks and benefits of their treatment recommendations. One option that must always be included is recommend that there not be any intervention. Another consideration, based on the patient–dentist interaction, particular needs/desires of the patient, and/or the skill/comfort level of the dentist, is to recommend referral to another practitioner. With regard to operative dentistry procedures, the decision to recommend surgical or nonsurgical intervention depends on the determination that a tooth is diseased, a restoration is defective, or the tooth or restoration is at some increased risk of further deterioration if the intervention does not occur. If any of these conditions exists, intervention is recommended to the patient. There may be multiple possible means by which to resolve the diagnosed disease or defect. Identification of treatment alternatives involves establishing a list of one or more reasonable interventions from the set of possible alternatives. Treatment alternatives for a specific condition may include, for example, periodic reevaluation to monitor the condition, chemotherapeutics (e.g., applications of fluoride to promote remineralization or antimicrobials to reduce bacteria), recontouring defective restorations or irregular tooth surfaces, repair of an existing restoration, and restoration of caries lesions or other defects. The list of reasonable treatment alternatives is based on current evidence of the effectiveness of treatments, prevailing standards of care, and clinical and nonclinical patient factors. If the decision is made to recommend intervention then identification and selection among treatment alternatives, with the patient's involvement, enables creation of the *treatment plan*.

The treatment plan is a carefully sequenced series of services designed to eliminate or control etiologic factors, repair existing damage, and create a functional, maintainable environment. An appropriate treatment plan depends on thorough evaluation of the patient, the expertise of the dentist, and a prediction of the patient's response to treatment. The treatment plan will also include strategies designed to reduce the patient's risk for future caries or other oral disease. Selection of specific components of the treatment plan is accomplished in consultation with the patient. The patient is advised of the reasonable treatment alternatives and related risks and benefits. After the patient is fully informed, the dentist and patient select a course of action that is most appropriate.

Treatment plans are influenced by many factors, including patient preferences, motivation, systemic health, emotional status, and financial resources. The treatment plan is also influenced by the dentist's knowledge, experience, and training; laboratory support; dentist–patient compatibility; availability of specialists; and the patient's functional, esthetic, and technical demands. Finally, a treatment plan is not a static list of services. Rather, it is often a multiphase and dynamic series of activities. Success of the treatment plan is determined by its ability to meet the patient's initial and long-term needs. A treatment plan should allow for reevaluation and be adaptable to meet the changing needs, preferences, and health conditions of the patient.

In the context of planning dental treatment, the clinician should recommend invasive operative treatment only when the benefits outweigh the risks of adverse outcomes. Restorations that require permanent removal of tooth structure have a limited lifespan. Studies have shown that the average lifespan of a restoration ranges from 5 to more than 15 years.[38] When the restoration is subsequently replaced additional tooth structure is removed, regardless of how carefully the operator removes the existing restoration. This situation results in what has been termed *the cycle of re-restoration*, which leads to larger and more invasive restorations over the course of a patient's life.[39]

As a general rule, remineralization therapies, as well as sealants in the case of pits and fissures, are the preferred methods of managing coronal lesions that are neither cavitated nor involve dentin. Remineralization is also recommended for root-surface lesions in which a break in the surface contour of the exposed root surface has not occurred. However, it is very important to note that *remineralization requires a high level of patient compliance with the therapeutic regimen and frequent recall visits to assess the success of the treatment*. If lesion progression is detected at recall, then operative intervention is warranted.

There are exceptions to the general rule of managing noncavitated enamel lesions with remineralization. Remineralization requires a shift in the delicate balance of the oral biofilm and therefore depends heavily on changes in patient behavior (e.g., improved home care, diet) and the timely application of antimicrobial agents, fluoride, and other remineralizing agents. Thus, when it is clear that the patient is unwilling or unable to follow the prescribed remineralization regimen of home care and professional care, it is often appropriate to remove the lesion(s) surgically and restore the defect or to seek to arrest the lesion (see Chapter 2).

If confirmed cavitation of the enamel or demineralization penetrating into the dentin on coronal surfaces is present or a break exists in the contour of exposed root and softening of the surface, then operative treatment is usually recommended. One exception to this general guideline is the lesion that is deemed arrested.

Treatment Plan Sequencing/Phasing

Proper sequencing is a crucial component of a successful treatment plan. Certain treatments must follow others in a logical order, whereas other treatments may or must occur concurrently and require coordination. Complex treatment plans often are sequenced in phases, including an urgent phase, a control phase, a reevaluation phase, a definitive phase, and a maintenance phase (that includes reassessment and recare). For most patients, the first three phases are accomplished simultaneously. Generally, the principle of "greatest need" guides the order in which treatment is sequenced. This principle suggests that what the patient needs most is performed first—with pain, bleeding, and swelling at the beginning of the treatment plan and elective esthetic procedures at the end. The process of treatment planning requires that the dentist develop an ever-increasing, comprehensive knowledge of dental disease management in the context of individualized patient care. Study of textbooks devoted to this discipline is indicated.[40]

Urgent Phase

The urgent phase of care begins with a thorough review of the patient's medical history and current condition. A patient presenting with swelling, pain, bleeding, or infection should have these problems managed as soon as possible, before initiation of subsequent phases.

Control Phase

A control phase is appropriate when the patient presents with multiple pressing problems and extensive active disease or when the prognosis is unclear. The goals of this phase are to remove etiologic factors, eliminate the ecologic niches of pathogens, and stabilize the patient's dental health. These goals are accomplished by (1) removal of active disease such that inflammation may resolve, (2) correction of conditions that prevent or limit hygiene efforts, (3) elimination of potential causes of disease, and (4) initiation of preventive activities. Examples of control phase treatment include extractions, endodontics, periodontal debridement and scaling, occlusal adjustment, caries arrest and/or removal, replacement or repair of defective restorations such as those with gingival overhangs, and use of caries control measures, as discussed in Chapter 2.

The dentist should develop a plan for the management and prevention of dental caries as part of the control phase. After the patient's caries status and caries risk have been determined, chemical, surgical, behavioral, mechanical, and dietary techniques may be used to improve host resistance and alter the oral flora.[40] Chapter 2 presents a detailed discussion of caries diagnosis, prevention, treatment, and control.

Reevaluation Phase

The reevaluation phase allows time between the control and definitive phases for resolution of inflammation and healing. Initial treatment and pulpal responses are reevaluated during this phase as the relative effectiveness of control phase treatment may influence and modify the definitive phase treatment plan. This phase is used to reinforce home care habits and assess motivation for further treatment.

Patients with an overall low risk profile, who only require minor alterations in diet, behaviors, and exposure to remineralization agents, may not require a formal control phase/reevaluation phase process. The treatment plan for these patients may start with a plan to definitively address immediate concerns while simultaneously implementing minor changes and reinforcing habits consistent with dental health.

Definitive Phase

The patient enters the definitive phase of treatment only after the dentist reassesses initial efforts to control disease and, with the patient, determines the need for further care. This phase may include endodontic, periodontal, orthodontic, and surgical procedures. The patient's active disease must be under control, and preventive efforts habitually established, before fixed or removable prosthodontic treatment. This phase is discussed in more detail in the section on interdisciplinary considerations in operative treatment planning.

Maintenance (Reassessment and Recare) Phase

The maintenance phase includes regular reassessment (synonyms include reevaluation, periodic) examinations that may reveal the need for adjustments to prevent future breakdown, provide an opportunity to reinforce home care, and plan recare treatment steps where disease has returned. Examinations for reassessment most frequently occur as part of strategically planned (recall) appointments for biofilm removal (dental prophylaxis). The frequency of reevaluation examinations depends, in large part, on the patient's risk for dental disease. A patient with a low risk profile may have longer intervals (e.g., 9–12 months) between recall visits. In contrast, patients at high risk profile should be recalled and examined much more frequently (e.g., 3–4 months).

Interdisciplinary Considerations in Operative Treatment Planning

When an operative procedure is performed during the control or definitive phases, general guidelines help determine when the operative treatment should occur relative to other forms of care. Following is a discussion on sequencing operative care with endodontic, periodontal, orthodontic, surgical, and prosthodontic treatments.

Endodontics

All teeth to be restored with large restorations should have a pulpal and periapical evaluation. If indicated, teeth should have endodontic treatment before restoration is completed. Also, a tooth previously endodontically treated, that shows no evidence of healing or has an inadequate filling or a filling exposed to oral fluids, should be evaluated for retreatment before restorative therapy is initiated.[41]

Periodontics

Generally, periodontal treatment should precede operative care, especially when improved oral hygiene and initial scaling/root planing procedures create (through reduction of gingival inflammation) a more desirable environment for performing operative treatment. A tooth with a questionable periodontal prognosis should not receive an extensive restoration until periodontal treatment provides a more favorable prognosis. If a tooth has a good periodontal prognosis, then operative treatment may occur before or after periodontal therapy, as long as the operative treatment is not compromised by the existing tissue condition. Treatment of deep caries lesions often requires caries control (see Chapter 2). Caries control may utilize temporization, creation of a foundation, or root canal therapy/foundation before periodontal therapy. The correction of gross restorative defects in restoration contours (e.g., open contact resulting from restoration undercontour, gingival overhang, poor embrasure form, occlusal interference resulting in increased mobility) is considered a part of initial periodontal therapy, and such corrections enable a more favorable tissue response. If periodontal surgical procedures are required, indirect restorations such as inlays or onlays, crowns, and prostheses should be delayed until the surgical phase is completed. Teeth planned for cast restorations may, however, be prepared and temporized before periodontal surgery. This approach permits confirmation of the restored tooth prognosis before surgery and allows improved access for the surgical procedure.

Patients with gingivitis and early periodontitis generally respond favorably to improved oral hygiene and scaling/root planing procedures. Patients with more advanced periodontitis might require removal (or at least minimization) of associated risk factors/indicators through surgical steps that eliminate/reduce sulcular depths or various regenerative procedures to resolve their periodontal disease. Steps to increase the zones of attached gingiva and eliminate abnormal frenal tension should be achieved by corrective periodontal surgical procedures around teeth receiving restorations with subgingival margins. In addition, any teeth requiring restorations that may encroach on the biologic width of the periodontium should

have appropriate crown-lengthening surgical procedures performed before the final restoration is placed. Usually, a minimum of 6 weeks is required after the surgery before final restorative procedures are undertaken.

Orthodontics

Orthodontic therapy, such as realignment or extrusion, may be required to provide improved interdental spacing, stress distribution, function, and esthetics. All caries lesions should be corrected with amalgam or composite restorations before orthodontic treatment begins. Few indications exist for cast restorations before orthodontic treatment is completed. In addition, patients undergoing orthodontic treatment should receive more intense focus (especially by the orthodontist) on the minimization/elimination of risk factors for caries and gingival/periodontal disease. The orthodontic treatment plan should include shorter recall intervals for biofilm removal, examination, and oral hygiene reinforcement.

Oral Surgery

In most instances, impacted, unerupted, and/or hopelessly diseased teeth should be removed before operative treatment. Oral surgery procedural steps required for third molar removal may jeopardize new restorations placed on second molars. In addition, soft tissue lesions, complicating exostoses, and improperly contoured ridge areas should be eliminated or corrected before final restorative care.

Fixed, Removable, and Implant Prosthodontics

Direct restorations should be completed, if possible, before placing indirect restorations. Large amalgam or composite foundation restorations must have secondary retention features (grooves, slots, pins) placed further from the external surface of the tooth so that the retention of the foundation material is not compromised during preparation for the indirect restoration. In removable prosthodontics, tooth preparations and restorations should allow for the design of the removable partial denture. This includes allowance for rests, guide planes, and clasps. The design of the direct restoration and the selection of appropriate restorative materials must be compatible with the design of the contemplated removable prosthesis. In cases where dental implants have been or will be placed, direct restorations should be planned and executed to allow necessary mesial, distal, and vertical (occlusal) space for implant-supported indirect restorations. Implant restorations may sometimes have unusual proximal contours, and adjacent amalgam or composite restorations should be designed to allow the best possible proximal contact relationships.

Treatment of Abrasion, Erosion, Abfraction, and Attrition

Abraded or eroded areas should be considered for restoration only if one or more of the following is true: (1) The area is affected by caries, (2) the defect is sufficiently deep to compromise the structural integrity of the tooth, (3) intolerable sensitivity exists and is unresponsive to conservative desensitizing measures, (4) the defect contributes to a gingival or periodontal problem (chronic biofilm accumulation), (5) the area is to be involved in the design of a removable partial denture, (6) the depth of the defect is such that there is increased risk of pulpal involvement, (7) the defect is actively progressing, or (8) the patient desires esthetic improvement. Areas of significant occlusal attrition that have exposed dentin, are sensitive, or annoying should be considered for restoration or at least protection from additional loss of tooth structure. The creation of an occlusal guard, for nocturnal use, may be indicated with a diagnosis of sleep-related bruxism. A treatment plan for definitive indirect restorations must include an occlusal analysis (which requires articulated diagnostic models) as part of the comprehensive examination. Careful consideration of related information from the patient assessment and examination process is essential if all aspects of the etiology are to be identified and risk factors reduced. Also, occlusal guard therapy should be considered for nocturnal protection of indirect restorations completed as part of the definitive phase.

Treatment of Root-Surface Caries

Root caries is common in older adults and in patients who have had periodontal therapy. Increases in the number of older patients in the patient population and tooth retention have contributed to this growing problem. Areas with root-surface caries usually should be restored when clinical and/or radiographic evidence of cavitation exists. Care must be exercised, however, to distinguish the active from the arrested (inactive) root-surface lesion. The arrested root-surface lesion may have sclerotic dentin that has darkened from extrinsic staining, is firm to the touch of an explorer, may be rough but is cleanable. Successful caries arrest usually occurs in patients whose oral hygiene or diet has improved such that the balance between demineralization and remineralization has become favorable. Generally, these lesions should not be restored except when the patient expresses esthetic concerns. If it is determined that the lesion needs restoration, it may be restored with tooth-colored materials or amalgam, depending on demands of the restorative material, preferences of the patient, and caries risk.

Prevention is preferred over restoration. It is recommended that appropriate preventive steps, such as improvement in diet/oral hygiene and fluoride treatment (with or without cementoplasty/dentinoplasty to eliminate surface roughness), be taken so as to limit carious breakdown and the need for restoration.

Treatment of Root-Surface Sensitivity

It is not unusual for patients to complain of root-surface sensitivity, which is an annoying sharp pain usually associated with gingival recession and exposed root surfaces. The most widely accepted explanation of this phenomenon is the *hydrodynamic theory*. This theory postulates that rapid dentinal tubule fluid movement toward the external surface of the tooth elongates odontoblastic processes (which extend from the pulp through the predentin and into dentin) and associated afferent nerve fibers. The elongation of the nerve fibers results in depolarization and the perception of pain (see Chapter 1). Causes of such fluid shifts include temperature changes, air-drying, and extreme osmotic gradients. Treatment methods that reduce rapid fluid shifts, by partially or totally occluding the ends of the exposed dentinal tubules, may help reduce the perceived sensitivity.

Dentinal hypersensitivity may become a problem when periodontal surgery causes clinical exposure of root surfaces (such that dentinal tubules are exposed and open). Numerous forms of nonsurgical treatment, such as fluoride varnishes, oxalate solutions, glutaraldehyde/HEMA-based desensitizers, resin-based adhesives, sealants, and desensitizing toothpastes that contain potassium nitrate, have been used to occlude the open tubules and, thereby, provide relief. When nonsurgical methods fail to provide relief, direct restorative treatment that physically covers the exposed dentin is indicated.

Treatment by Repair and Recontour of Existing Restorations

Amalgam, composite, or indirect restorations often may be repaired or recontoured as opposed to completely removed and replaced. Growing evidence suggests that the removal and replacement of restorations result in the cycle of re-restoration, which leads to increasingly larger tooth preparations and the resultant trauma to the tooth and supporting structures.[39] In addition, resurfacing or repair of composites and repair of cast restorations has been shown to be effective.[42-44] Also, amalgam restorations with localized defects may be repaired with amalgam or with sealant resins.[17,42] If a restoration has an isolated carious defect, and complete removal of the caries lesion has been confirmed, then it is acceptable and often preferable to restore the isolated area without replacement of the whole restoration. In many instances, recontouring or resurfacing the existing restoration may delay replacement and is an acceptable form of treatment.

Treatment by Replacement of Existing Restorations

Indications for replacing restorations include the following: (1) marginal void(s), especially in the gingival one third, that cannot be repaired and predispose to caries formation; (2) poor proximal contour or a gingival overhang that contributes to periodontal breakdown; (3) a marginal ridge discrepancy that contributes to food impaction; (4) overcontouring of a facial or lingual surface resulting in biofilm accumulation gingival to the height of contour and resultant inflammation of gingiva overprotected from the cleansing action of food bolus or toothbrush; (5) poor proximal contact that is either open or improper in location or size, resulting in interproximal food impaction and inflammation of impacted gingival papilla; (6) recurrent caries that cannot be treated adequately by a repair restoration; and (7) superficial marginal gap formation (ditching) deeper than 0.5 mm that predisposes to caries.[44]

Indications for replacing tooth-colored restorations include (1) improper contours that cannot be repaired, (2) large voids, (3) deep marginal staining, (4) recurrent caries, and (5) unacceptable esthetics.[44] Bonded restorations that have superficial marginal staining may be corrected by shallow, narrow, marginal repair.

Treatment With Amalgam Restorations

Dental amalgam still is recognized as one of the most successful direct restorative materials and is especially indicated for patients deemed to be moderate or high caries risk.[45] Inaccurate information with regard to the safety of amalgam has resulted in controversy among health care providers, environmentalists, legislators, and the general population. Although the use of amalgam is considered safe by multiple independent agencies, the release of elemental mercury does contribute to environmental levels. Therefore responsible handling is important. Chapter 10 presents the current indications for amalgam restorations and Chapter 13 presents a more complete discussion of legitimate mercury concerns and the safe use of dental amalgam.

Treatment With Direct Composite and Other Tooth-Colored Restorations

Direct composite restorations are indicated for the treatment of many lesions in anterior and posterior teeth. Successful treatment outcomes require meticulous attention to detail with regard to the enamel/dentin substrate and the properties of the specific adhesive system/composite resin being used. Correct application will result in the rewarding creation of form, function, and lifelike appearance of missing tooth structure. Determination of patient caries risk is important when considering the use of composite resin-based restorations. A systematic review has identified that the likelihood of development of recurrent caries adjacent to composite resin restorations is at least twice that of amalgam restorations in high caries risk patients.[45] Detailed indications for composite and other tooth-colored restorations are presented in Chapter 8.

Treatment With Indirect Cast-Metal Restorations

Partial- or full-coverage indirect cast-metal (primarily gold alloy) onlay restorations still remain among the most predictable and dependable restorations available to dentistry. These are the conservative restoration of choice for compromised teeth in high stress areas. The benefit of these restorations is that they cover and reinforce cusps without removal of healthy tooth structure in the middle and cervical areas of the facial and lingual surfaces (see Online Chapter 18). Indirect cast-metal restoration of the total clinical crown of teeth allows complete control of all contours and, thereby, the creation of anatomic shape consistent with optimal occlusal function and gingival health.

Treatment With Indirect Tooth-Colored Restorations

Properly designed porcelain-fused-to-metal (PFM) indirect restorations have clinically proven, long-term success in the restoration of individual teeth and edentulous areas. The use of ceramic materials without metal substrates has steadily increased in recent years. Partial-coverage bonded indirect tooth-colored restorations may be indicated for the restoration of large defects in low stress areas when esthetics and optimal control of contours is necessary. Full-coverage bonded indirect tooth-colored restorations also may be selected for the conservative restoration of weakened posterior teeth in low stress, esthetically critical areas.

The use of tooth-colored, zirconia-based, indirect restorations has steadily increased over the last two decades. *In vitro* and short-term *in vivo* research studies suggest that the clinical durability of these crowns may allow use as an alternative to indirect cast-metal or PFM restorations. However, there are currently no published long-term randomized, controlled clinical trials verifying this to actually be the case. These restorations may be generated through the use of traditional impression/dental laboratory techniques or through the use of computer-assisted digital impression, design and manufacturing processes. Indirect tooth-colored restorations are covered in more detail in Chapter 12.

Treatment of Esthetic Concerns

Interest in smile esthetics is growing among many segments of the population. As a result, a range of treatments has been developed to manage a wide array of esthetic concerns. Chapter 9 describes conservative esthetic treatments, which include selective recontouring of anterior teeth, vital bleaching, and microabrasion. These conservative approaches have well-documented outcomes. In addition to these conservative techniques, advances in direct composite restorations have permitted the closure of diastemas, recontouring of

teeth, and other tooth additions by means other than extensive full-coverage indirect restorations.

Treatment Considerations for Older Patients

In the past, older adults constituted a relatively minor proportion of the population. Older individuals used dental services infrequently because most were edentulous, had limited financial resources, and delayed unmet dental needs until they became symptomatic. Today, individuals 65 years and older represent a rapidly growing segment of the population. Older individuals now are better educated consumers, have greater financial resources, are more prevention minded, and have retained more teeth as compared with their predecessors. However, older individuals are living with increasingly more complex medical, mental, emotional, and social conditions that affect their ability to care for their dentition and periodontium. These conditions must be considered when planning dental treatment. Financial and social barriers prevent some older individuals from seeking oral health care. Although, as a group, older adults enjoy greater financial resources, many remain on restricted budgets and are faced with tough decisions regarding the spending of limited resources. Transportation to and from the dental office becomes complicated for those who no longer drive. A comprehensive review of geriatric dentistry is beyond the scope of this chapter; rather, some issues that are important for treatment planning for older patients are highlighted.[46-48]

Clear and effective communication is crucial. Many older adults have hearing loss and dentists must speak more distinctly and at a higher volume. Patients with memory loss appreciate written summaries and instructions that assist them in remembering details of the visit and planned treatment when they leave the dental office. The use of large simple fonts in written communications is particularly helpful to patients with diminished visual acuity.

An accurate medical history, risk assessment, and integration of dental and medical care are particularly important considerations for older patients. Many chronic diseases of the cardiovascular, respiratory, endocrine, renal, gastrointestinal, musculoskeletal, immune, and neurologic systems are associated with aging, influence dental disease, and complicate dental treatment decision making. Cardiovascular disease, Alzheimer disease, depression, osteoarthritis, rheumatoid arthritis, osteoporosis, cancer, and diabetes are a few of the diseases that commonly affect older adults, and their medical management increases in complexity with advancing years. It is estimated that older individuals living in community settings take an average of four medications each day; six of the top 10 drugs prescribed in 2001 were used to treat age-related chronic conditions.[46,47] Many of these medications have the potential for adverse drug reactions and drug interactions. Oral adverse effects include dry mouth (xerostomia), increased bleeding of tissues, lichenoid reactions, tissue overgrowth, and hypersensitivity reactions. The dentist must be aware of the impact these medications may have on dental treatment planning and management. Consultation with the patient's physician is highly recommended so as to gain understanding of these medical, mental, and emotional conditions and their potential impact on dental treatment. The dentist should recognize the impact of polypharmacy on salivary flow, especially the use of xerostomic medications, and discuss with the physician the potential substitution of medications with fewer xerostomic effects.

Oral changes associated with undernourishment, immunosuppression, dehydration, smoking, alcohol use, disease, medications, and dental problems lead to a depressed sense of taste and smell in older patients.[49] Perceptions of salty and bitter tastes and olfactory function decline with age, whereas perceptions of sweet and sour tastes do not. As a result, food may become tasteless and unappetizing, and more sugars, fats, and salts are added in an attempt to increase flavor. Undernourished individuals are encouraged to consume calorie-rich, complete-nutrition beverages, which also are rich in refined carbohydrates. Smoking reduces the taste of foods by causing physical coating of the tongue and regression of the taste buds on the tongue and olfactory receptors in the roof of the nasal cavity over time. Inadequate fluid intake may lead to chronic dehydration and altered taste perception. These practices increase the risk of dental disease in this population. Dietary assessment and counseling are crucial in older patients to identify inadequate diets and suggest modifications that enhance taste and smell while lowering the risks of dental disease. Herb seasonings may enhance the flavor of foods in lieu of sugar and salt. Salivary stimulants, citrus-flavored candies containing xylitol or other sugar replacements, tongue brushing or scraping, and smoking cessation are some additional measures that may promote taste and olfactory perception in older adults.

Dental and periodontal diseases may progress more rapidly in older adults.[47] Dental caries, particularly root caries, is the most significant reason for tooth loss in older adults. Ineffective plaque removal, xerostomia, soft sugar-rich diets, fixed and removable prostheses, abrasions at the CEJ, gingival recession, and chronic periodontal inflammation (with increased activity of collagenolytic enzymes) make root surfaces more prone to caries compared with other surfaces. Root-surface restorations are challenging to successfully perform and are at risk of recurrent decay in the future. Careful selection of restoration design, materials, and finishing is essential if the patient is to be able to perform successful biofilm removal and thereby maximize the longevity of restorations. Also, many dental practitioners prefer to intervene more aggressively with dental treatment rather than take a "watchful waiting" approach. As more teeth are being retained and have large restorations at risk of fracture or recurrent decay, attention must be placed on developing cost-effective and innovative means of restoring teeth, particularly for older individuals on a limited budget. The cost-effective use of silver diamine fluoride (SDF) to arrest caries, even though the treated area becomes darkly stained, may be an optimal treatment option in this population (see Chapter 2).

Prevention of dental disease increases in importance but becomes more challenging in older adults. Physical limitations such as arthritis, Parkinson disease, vision impairment, and other chronic illnesses reduce the patients' ability to clean their teeth and periodontal tissues effectively. Powered rotation–oscillation toothbrushes and manual toothbrushes with larger handles, for easier gripping, are recommended for patients with decreased manual dexterity. Consistent use of fluoride-containing dentifrices and other remineralization products, antimicrobial mouthrinses, oral pH management, flossing, oral irrigation, and chewing of xylitol gum may reduce the risk of developing dental caries and periodontal infection.[50] Written reminders are useful to serve as aids for older patients who forget to brush their teeth because of memory loss associated with Alzheimer disease. Because many older individuals may have never been taught how to effectively clean their teeth, the dentist must observe their technique and instruct them in proper oral hygiene procedures to be performed after each meal. Dentists must carefully inform patients in the proper application method of 5000-ppm fluoride toothpastes. Incorrect application (e.g., rinsing/eating/drinking immediately after brushing) severely limits any potential benefit.

A unique aspect of aging is an increasing reliance on the assistance of caregivers with activities of daily living. As a result, the dentist must work with caregivers who provide dental care for patients in the home, assisted living facility, nursing home, and hospital settings. The dental professional may need to spend more time educating and training the caregiver, rather than the patient, in the importance of oral hygiene and effective plaque removal techniques.

Treatment Plan Approval

Informing patients well about their conditions and treatment options and then obtaining their consent has become an integral part of contemporary dental practice.[51] One aspect of informed consent is to provide patients with the necessary information about the alternative therapies available to manage their oral conditions. For nearly all conditions, usually more than one treatment alternative is available. These alternatives, with their advantages and disadvantages, should be presented to the patient. In addition, the patient should be informed of the risks associated with each alternative therapy. Dentists must remember that a reasonable alternative often is *not* to intervene directly with restorative care. Rather, based on the nature of dental disease progression, elimination or reduction of risk factors/indicators may need to be the initial focus while monitoring the condition. Even these intentional efforts are part of a treatment plan and must be included in the informed consent process. Proactive conservative steps, in the case of caries, may be to attempt to remineralize or arrest the lesion(s). Finally, the cost of treatment alternatives should be discussed with the patient. Treatment may proceed when the dentist is sure that the patient has a full and complete understanding of the alternative treatments, their associated risks and benefits, and the results of possible nontreatment.[51-53]

Summary

Proper assessment, examination, diagnosis, and treatment planning play a crucial role in the delivery of quality dental care. Each patient must be evaluated (examined) individually in a thorough and systematic fashion. After the patient's preferences, risks, and condition(s) are understood and recorded, a treatment plan may be developed and implemented. A successful treatment plan carefully sequences and integrates all necessary procedures indicated for the patient. Few absolutes exist in treatment planning; the available information must be considered carefully and incorporated into a sequenced approach that fits the desires/needs of the individual. Patients must have an active role in the process; they must be informed of the findings, advised of the risks and benefits of proposed treatment, and given the opportunity to decide the course of treatment. The process of patient assessment, examination, diagnosis, and treatment planning represents one of the greatest challenges in dentistry and is rewarding for both the patient and the dentist if done properly (i.e., thoroughly and with the patient's best interests in mind).

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